tropical: ConvexHull

2026-10-26T03:17:59

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `aff15420a136`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6112	[0.608, 0.614]	0.0044	5	0
adaptive	pointer	0.01	0.6107	[0.609, 0.612]	0.00211	5	0
adaptive	pointer	0.1	0.6102	[0.609, 0.612]	0.00201	5	0
adaptive	pointer	1	0.6097	[0.608, 0.611]	0.00159	5	0
adaptive	pointer	10	0.6099	[0.609, 0.611]	0.0016	5	0
adaptive	pointer	100	0.61	[0.608, 0.612]	0.00263	5	0
adaptive	pointer	1000	0.6096	[0.608, 0.612]	0.00264	5	0
adaptive	pointer	1e+04	0.6055	[0.603, 0.608]	0.00355	5	0
adaptive	seq2seq	0	0.7155	[0.711, 0.72]	0.00602	5	0
adaptive	seq2seq	0.01	0.7199	[0.713, 0.728]	0.00931	5	0
adaptive	seq2seq	0.1	0.7201	[0.714, 0.728]	0.00926	5	0
adaptive	seq2seq	1	0.7189	[0.712, 0.726]	0.00869	5	0
adaptive	seq2seq	10	0.7198	[0.714, 0.727]	0.00822	5	0
adaptive	seq2seq	100	0.7188	[0.712, 0.727]	0.00893	5	0
adaptive	seq2seq	1000	0.7004	[0.69, 0.71]	0.0131	5	0
adaptive	seq2seq	1e+04	0.6824	[0.669, 0.696]	0.0174	5	0
tropical	pointer	0	0.785	[0.763, 0.803]	0.0271	5	0
tropical	pointer	0.01	0.7947	[0.771, 0.819]	0.0302	5	0
tropical	pointer	0.1	0.7924	[0.769, 0.821]	0.0335	5	0
tropical	pointer	1	0.7818	[0.766, 0.799]	0.0231	5	0
tropical	pointer	10	0.797	[0.779, 0.824]	0.0305	5	0
tropical	pointer	100	0.7871	[0.78, 0.795]	0.0101	5	0
tropical	pointer	1000	0.7873	[0.771, 0.803]	0.022	5	0
tropical	pointer	1e+04	0.7964	[0.787, 0.805]	0.0112	5	0
tropical	seq2seq	0	0.9648	[0.958, 0.97]	0.0078	5	0
tropical	seq2seq	0.01	0.9602	[0.951, 0.969]	0.012	5	0
tropical	seq2seq	0.1	0.964	[0.957, 0.972]	0.0102	5	0
tropical	seq2seq	1	0.9651	[0.953, 0.973]	0.0136	5	0
tropical	seq2seq	10	0.9649	[0.958, 0.973]	0.0101	5	0
tropical	seq2seq	100	0.9613	[0.95, 0.973]	0.015	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.93	[0.917, 0.94]	0.0141	5	0
tropical	seq2seq	1e+04	0.7699	[0.732, 0.807]	0.0501	5	0
vanilla	pointer	0	0.5794	[0.574, 0.585]	0.00748	5	0
vanilla	pointer	0.01	0.5792	[0.574, 0.587]	0.0086	5	0
vanilla	pointer	0.1	0.5789	[0.574, 0.586]	0.00795	5	0
vanilla	pointer	1	0.5791	[0.574, 0.587]	0.00842	5	0
vanilla	pointer	10	0.5789	[0.574, 0.586]	0.00827	5	0
vanilla	pointer	100	0.579	[0.574, 0.587]	0.00879	5	0
vanilla	pointer	1000	0.5793	[0.574, 0.588]	0.00945	5	0
vanilla	pointer	1e+04	0.5783	[0.572, 0.585]	0.00838	5	0
vanilla	seq2seq	0	0.6728	[0.667, 0.681]	0.00953	5	0
vanilla	seq2seq	0.01	0.6687	[0.661, 0.676]	0.01	5	0
vanilla	seq2seq	0.1	0.6691	[0.662, 0.676]	0.00946	5	0
vanilla	seq2seq	1	0.6691	[0.662, 0.676]	0.00966	5	0
vanilla	seq2seq	10	0.6736	[0.668, 0.679]	0.007	5	0
vanilla	seq2seq	100	0.672	[0.665, 0.681]	0.0101	5	0
vanilla	seq2seq	1000	0.6626	[0.653, 0.674]	0.0133	5	0
vanilla	seq2seq	1e+04	0.619	[0.61, 0.63]	0.013	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-4.625×10^{-4}	[-0.00522, 0.0043]	0.9999	
adaptive	pointer	0.1	-9.339×10^{-4}	[-0.0057, 0.00383]	0.9926	
adaptive	pointer	1	-0.001433	[-0.0062, 0.00333]	0.9325	
adaptive	pointer	10	-0.00124	[-0.006, 0.00352]	0.9660	
adaptive	pointer	100	-0.001202	[-0.00596, 0.00356]	0.9709	
adaptive	pointer	1000	-0.00157	[-0.00633, 0.00319]	0.8996	
adaptive	pointer	1e+04	-0.005681	[-0.0104, -9.19×10^{-4}]	0.0140	*
adaptive	seq2seq	0.01	0.004473	[-0.0141, 0.0231]	0.9774	
adaptive	seq2seq	0.1	0.004621	[-0.014, 0.0232]	0.9732	
adaptive	seq2seq	1	0.003468	[-0.0151, 0.0221]	0.9945	
adaptive	seq2seq	10	0.004375	[-0.0142, 0.023]	0.9798	
adaptive	seq2seq	100	0.003281	[-0.0153, 0.0219]	0.9960	
adaptive	seq2seq	1000	-0.01503	[-0.0336, 0.00357]	0.1540	
adaptive	seq2seq	1e+04	-0.03302	[-0.0516, -0.0144]	0.0002	*
tropical	pointer	0.01	0.00963	[-0.0338, 0.0531]	0.9854	
tropical	pointer	0.1	0.007389	[-0.0361, 0.0508]	0.9968	
tropical	pointer	1	-0.003213	[-0.0467, 0.0402]	1.0000	
tropical	pointer	10	0.01198	[-0.0315, 0.0554]	0.9551	
tropical	pointer	100	0.002062	[-0.0414, 0.0455]	1.0000	
tropical	pointer	1000	0.00227	[-0.0412, 0.0457]	1.0000	
tropical	pointer	1e+04	0.01139	[-0.0321, 0.0548]	0.9648	
tropical	seq2seq	0.01	-0.004555	[-0.0412, 0.0321]	0.9996	
tropical	seq2seq	0.1	-7.782×10^{-4}	[-0.0374, 0.0359]	1.0000	
tropical	seq2seq	1	3.253×10^{-4}	[-0.0363, 0.037]	1.0000	
tropical	seq2seq	10	1.233×10^{-4}	[-0.0365, 0.0368]	1.0000	
tropical	seq2seq	100	-0.003478	[-0.0401, 0.0332]	0.9999	
tropical	seq2seq	1000	-0.03478	[-0.0715, 0.00189]	0.0687	
tropical	seq2seq	1e+04	-0.1949	[-0.232, -0.158]	$< 10^{-4}$	*
vanilla	pointer	0.01	-2.280×10^{-4}	[-0.015, 0.0145]	1.0000	
vanilla	pointer	0.1	-5.537×10^{-4}	[-0.0153, 0.0142]	1.0000	
vanilla	pointer	1	-2.986×10^{-4}	[-0.015, 0.0144]	1.0000	

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
vanilla	pointer	10	-5.047×10^{-4}	[-0.0152, 0.0142]	1.0000	
vanilla	pointer	100	-4.481×10^{-4}	[-0.0152, 0.0143]	1.0000	
vanilla	pointer	1000	-1.444×10^{-4}	[-0.0149, 0.0146]	1.0000	
vanilla	pointer	1e+04	-0.001157	[-0.0159, 0.0136]	1.0000	
vanilla	seq2seq	0.01	-0.004168	[-0.0224, 0.0141]	0.9827	
vanilla	seq2seq	0.1	-0.003708	[-0.0219, 0.0145]	0.9909	
vanilla	seq2seq	1	-0.003778	[-0.022, 0.0145]	0.9899	
vanilla	seq2seq	10	7.772×10^{-4}	[-0.0175, 0.019]	1.0000	
vanilla	seq2seq	100	-8.447×10^{-4}	[-0.0191, 0.0174]	1.0000	
vanilla	seq2seq	1000	-0.01026	[-0.0285, 0.00798]	0.4821	
vanilla	seq2seq	1e+04	-0.05389	[-0.0721, -0.0357]	$< 10^{-4}$	*

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-5.964×10^{-4}	$[-8.24 \times 10^{-4}, -4.04 \times 10^{-4}]$	0.0076	-0.326	0.0144
adaptive	seq2seq	-0.005428	[-0.00711, -0.00375]	$< 10^{-4}$	-0.501	0.0002
tropical	pointer	1.169×10^{-5}	[-0.00302, 0.00303]	0.9953	0.131	0.3229
tropical	seq2seq	-0.02296	[-0.0269, -0.019]	$< 10^{-4}$	-0.512	0.0002
vanilla	pointer	-7.560×10^{-5}	$[-2.54 \times 10^{-4}, 1.20 \times 10^{-4}]$	0.9105	-0.0552	0.6897
vanilla	seq2seq	-0.00569	[-0.00744, -0.00393]	0.0004	-0.291	0.0266

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.2056	[0.167, 0.244]	0.0005	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.2155	[0.181, 0.25]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.2135	[0.171, 0.256]	0.0005	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.2027	[0.172, 0.233]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.2181	[0.175, 0.261]	0.0005	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.2081	[0.187, 0.229]	$< 10^{-4}$	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.208	[0.179, 0.237]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.2182	[0.204, 0.233]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.2919	[0.283, 0.301]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.2915	[0.265, 0.318]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.2948	[0.274, 0.316]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.296	[0.269, 0.323]	$< 10^{-4}$	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.2913	[0.277, 0.306]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.2893	[0.266, 0.312]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.2674	[0.245, 0.29]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.1509	[0.102, 0.2]	0.0010	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.03175	[0.0185, 0.045]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.03151	[0.0203, 0.0427]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.03137	[0.0214, 0.0413]	0.0094	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.03061	[0.0203, 0.041]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.03101	[0.0205, 0.0415]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.03099	[0.0195, 0.0425]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.03032	[0.018, 0.0427]	0.0108	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02722	[0.0152, 0.0392]	0.0108	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.04262	[0.0323, 0.053]	0.0040	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.05127	[0.0393, 0.0632]	0.0040	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.05095	[0.0388, 0.0632]	0.0040	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.04987	[0.038, 0.0617]	0.0040	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.04622	[0.0377, 0.0548]	0.0018	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.04675	[0.0363, 0.0572]	0.0036	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.03785	[0.025, 0.0507]	0.0108	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.06349	[0.0398, 0.0872]	0.0108	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1043	[-0.11, -0.0987]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1092	[-0.118, -0.1]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1098	[-0.119, -0.101]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1092	[-0.119, -0.0993]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1099	[-0.119, -0.101]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.1088	[-0.117, -0.1]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.09083	[-0.104, -0.0775]	0.0008	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.07695	[-0.0949, -0.0589]	0.0021	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.1797	[-0.213, -0.147]	0.0014	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.1655	[-0.202, -0.129]	0.0021	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.1716	[-0.212, -0.131]	0.0021	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.1833	[-0.208, -0.159]	0.0006	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.1679	[-0.214, -0.122]	0.0027	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.1742	[-0.202, -0.146]	0.0010	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.1427	[-0.182, -0.104]	0.0027	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.02655	[-0.0383, 0.0914]	0.3193	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.09341	[-0.108, -0.0786]	0.0010	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.08947	[-0.11, -0.0693]	0.0021	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.09026	[-0.109, -0.0718]	0.0020	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.08993	[-0.109, -0.0706]	0.0021	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.09469	[-0.111, -0.0788]	0.0011	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.09301	[-0.113, -0.0735]	0.0021	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0833	[-0.107, -0.0595]	0.0027	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.04068	[-0.059, -0.0224]	0.0070	*

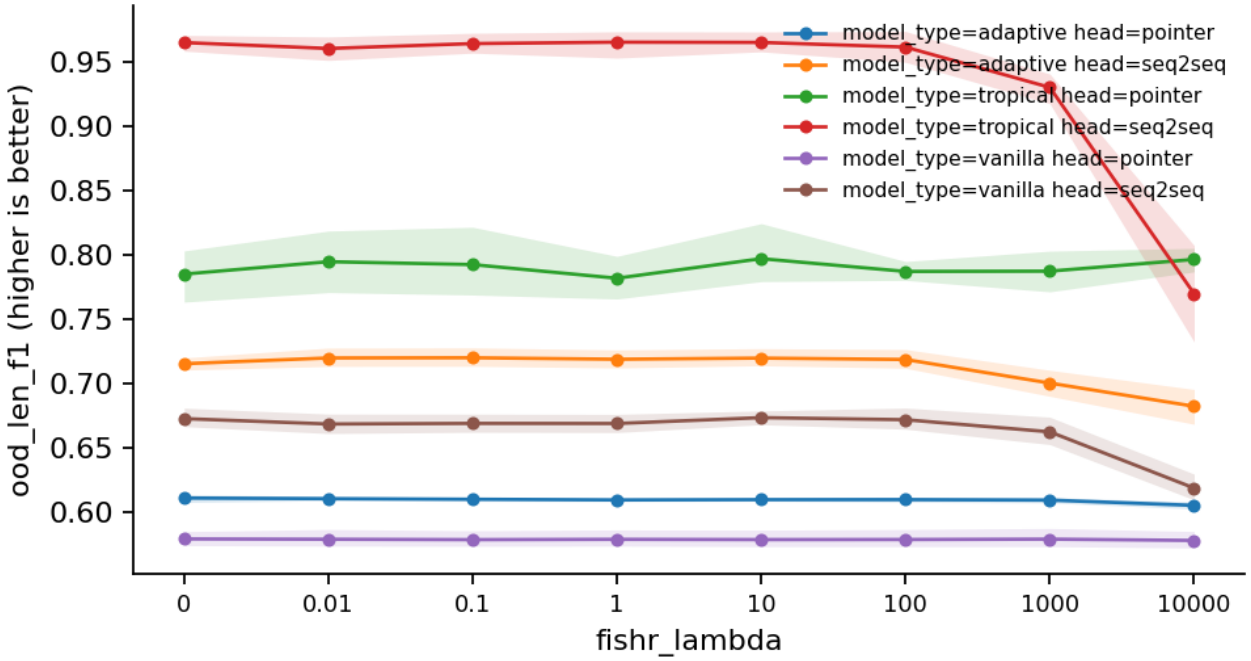


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

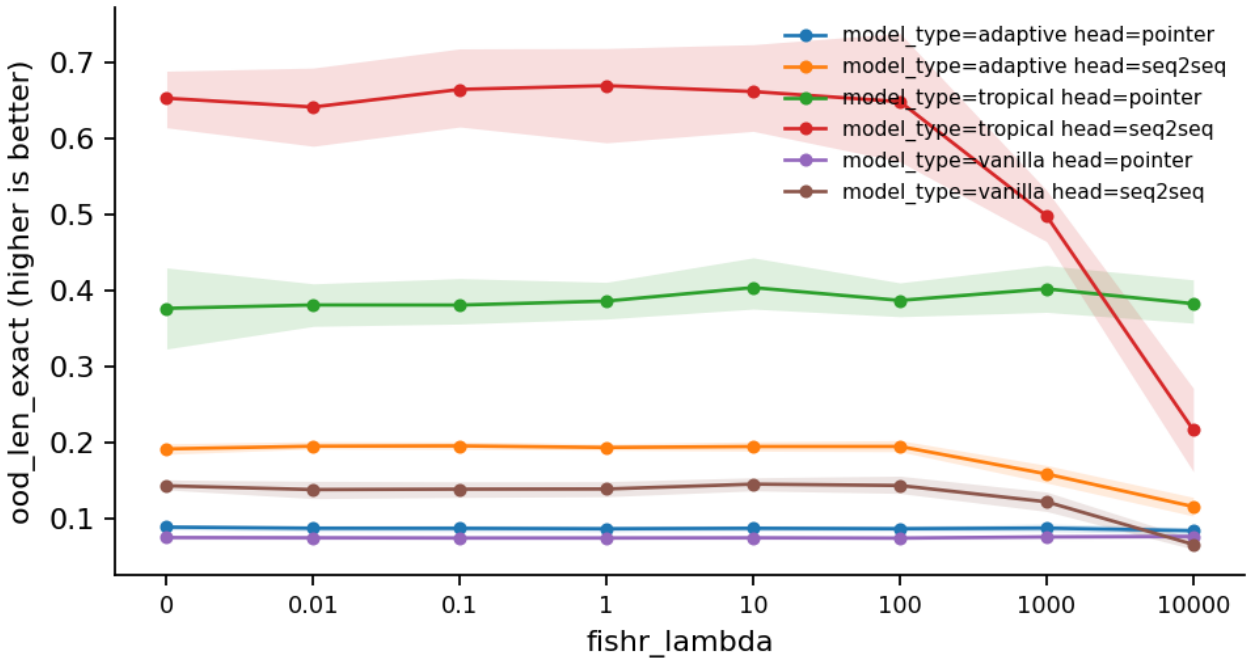


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.08807	[0.0847, 0.0914]	0.00475	5	0
adaptive	pointer	0.01	0.08653	[0.0825, 0.0905]	0.00527	5	0
adaptive	pointer	0.1	0.08643	[0.0823, 0.0905]	0.00537	5	0
adaptive	pointer	1	0.08587	[0.082, 0.0897]	0.00498	5	0
adaptive	pointer	10	0.0865	[0.0828, 0.0902]	0.00474	5	0
adaptive	pointer	100	0.08593	[0.0824, 0.0898]	0.00474	5	0
adaptive	pointer	1000	0.08673	[0.0823, 0.0924]	0.00634	5	0
adaptive	pointer	1e+04	0.0833	[0.0789, 0.0875]	0.00563	5	0
adaptive	seq2seq	0	0.1912	[0.184, 0.198]	0.00884	5	0
adaptive	seq2seq	0.01	0.1948	[0.191, 0.201]	0.00687	5	0
adaptive	seq2seq	0.1	0.1951	[0.191, 0.201]	0.00659	5	0
adaptive	seq2seq	1	0.1931	[0.19, 0.198]	0.00567	5	0
adaptive	seq2seq	10	0.1943	[0.188, 0.201]	0.0077	5	0
adaptive	seq2seq	100	0.1944	[0.188, 0.202]	0.00953	5	0
adaptive	seq2seq	1000	0.1581	[0.146, 0.17]	0.0152	5	0
adaptive	seq2seq	1e+04	0.1151	[0.103, 0.127]	0.0159	5	0
tropical	pointer	0	0.3766	[0.323, 0.43]	0.0697	5	0
tropical	pointer	0.01	0.381	[0.353, 0.409]	0.0364	5	0
tropical	pointer	0.1	0.3809	[0.356, 0.416]	0.039	5	0
tropical	pointer	1	0.3862	[0.363, 0.411]	0.0355	5	0
tropical	pointer	10	0.404	[0.376, 0.443]	0.0444	5	0
tropical	pointer	100	0.387	[0.366, 0.41]	0.0291	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.4023	[0.372, 0.433]	0.0409	5	0
tropical	pointer	1e+04	0.3827	[0.357, 0.414]	0.0384	5	0
tropical	seq2seq	0	0.6538	[0.615, 0.69]	0.0489	5	0
tropical	seq2seq	0.01	0.6421	[0.591, 0.694]	0.0703	5	0
tropical	seq2seq	0.1	0.6654	[0.616, 0.719]	0.0679	5	0
tropical	seq2seq	1	0.6703	[0.595, 0.719]	0.0827	5	0
tropical	seq2seq	10	0.6625	[0.61, 0.725]	0.0738	5	0
tropical	seq2seq	100	0.6492	[0.57, 0.739]	0.106	5	0
tropical	seq2seq	1000	0.4986	[0.465, 0.532]	0.0436	5	0
tropical	seq2seq	1e+04	0.2163	[0.162, 0.271]	0.0735	5	0
vanilla	pointer	0	0.07427	[0.0718, 0.0763]	0.00284	5	0
vanilla	pointer	0.01	0.074	[0.0711, 0.0763]	0.00335	5	0
vanilla	pointer	0.1	0.07373	[0.0712, 0.0759]	0.00309	5	0
vanilla	pointer	1	0.07373	[0.0712, 0.0763]	0.00314	5	0
vanilla	pointer	10	0.07397	[0.0713, 0.0761]	0.00321	5	0
vanilla	pointer	100	0.07357	[0.0712, 0.0761]	0.00296	5	0
vanilla	pointer	1000	0.0751	[0.0723, 0.0779]	0.00338	5	0
vanilla	pointer	1e+04	0.0758	[0.0726, 0.0793]	0.00462	5	0
vanilla	seq2seq	0	0.1425	[0.137, 0.15]	0.00886	5	0
vanilla	seq2seq	0.01	0.1375	[0.126, 0.149]	0.0149	5	0
vanilla	seq2seq	0.1	0.1381	[0.127, 0.148]	0.0136	5	0
vanilla	seq2seq	1	0.1383	[0.128, 0.148]	0.0136	5	0
vanilla	seq2seq	10	0.1447	[0.136, 0.153]	0.0113	5	0
vanilla	seq2seq	100	0.1429	[0.133, 0.155]	0.014	5	0
vanilla	seq2seq	1000	0.1214	[0.109, 0.134]	0.0164	5	0
vanilla	seq2seq	1e+04	0.06523	[0.059, 0.0727]	0.00864	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.001533	[-0.0107, 0.00764]	0.9971	
adaptive	pointer	0.1	-0.001633	[-0.0108, 0.00754]	0.9958	
adaptive	pointer	1	-0.0022	[-0.0114, 0.00698]	0.9777	
adaptive	pointer	10	-0.001567	[-0.0107, 0.00761]	0.9967	
adaptive	pointer	100	-0.002133	[-0.0113, 0.00704]	0.9810	
adaptive	pointer	1000	-0.001333	[-0.0105, 0.00784]	0.9988	
adaptive	pointer	1e+04	-0.004767	[-0.0139, 0.00441]	0.5635	
adaptive	seq2seq	0.01	0.0036	[-0.0142, 0.0214]	0.9913	
adaptive	seq2seq	0.1	0.003967	[-0.0139, 0.0218]	0.9851	
adaptive	seq2seq	1	0.001933	[-0.0159, 0.0198]	0.9998	
adaptive	seq2seq	10	0.003133	[-0.0147, 0.021]	0.9961	
adaptive	seq2seq	100	0.003233	[-0.0146, 0.0211]	0.9953	
adaptive	seq2seq	1000	-0.03307	[-0.0509, -0.0152]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.07603	[-0.0939, -0.0582]	$< 10^{-4}$	*
tropical	pointer	0.01	0.004433	[-0.071, 0.0799]	1.0000	
tropical	pointer	0.1	0.0043	[-0.0712, 0.0798]	1.0000	
tropical	pointer	1	0.009567	[-0.0659, 0.085]	0.9995	
tropical	pointer	10	0.02737	[-0.0481, 0.103]	0.8536	
tropical	pointer	100	0.0104	[-0.0651, 0.0859]	0.9991	
tropical	pointer	1000	0.0257	[-0.0498, 0.101]	0.8855	
tropical	pointer	1e+04	0.006133	[-0.0693, 0.0816]	1.0000	
tropical	seq2seq	0.01	-0.01167	[-0.139, 0.116]	0.9999	
tropical	seq2seq	0.1	0.0116	[-0.116, 0.139]	0.9999	

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	seq2seq	1	0.01653	[-0.111, 0.144]	0.9994	
tropical	seq2seq	10	0.008733	[-0.119, 0.136]	1.0000	
tropical	seq2seq	100	-0.004567	[-0.132, 0.123]	1.0000	
tropical	seq2seq	1000	-0.1552	[-0.283, -0.0276]	0.0118	*
tropical	seq2seq	1e+04	-0.4375	[-0.565, -0.31]	$< 10^{-4}$	*
vanilla	pointer	0.01	-2.667×10^{-4}	[-0.00614, 0.00561]	1.0000	
vanilla	pointer	0.1	-5.333×10^{-4}	[-0.00641, 0.00534]	0.9999	
vanilla	pointer	1	-5.333×10^{-4}	[-0.00641, 0.00534]	0.9999	
vanilla	pointer	10	-3.000×10^{-4}	[-0.00618, 0.00558]	1.0000	
vanilla	pointer	100	-7.000×10^{-4}	[-0.00658, 0.00518]	0.9997	
vanilla	pointer	1000	8.333×10^{-4}	[-0.00504, 0.00671]	0.9990	
vanilla	pointer	1e+04	0.001533	[-0.00434, 0.00741]	0.9657	
vanilla	seq2seq	0.01	-0.005033	[-0.0276, 0.0176]	0.9849	
vanilla	seq2seq	0.1	-0.004467	[-0.0271, 0.0181]	0.9923	
vanilla	seq2seq	1	-0.004233	[-0.0268, 0.0184]	0.9943	
vanilla	seq2seq	10	0.0022	[-0.0204, 0.0248]	0.9999	
vanilla	seq2seq	100	4.000×10^{-4}	[-0.0222, 0.023]	1.0000	
vanilla	seq2seq	1000	-0.02117	[-0.0438, 0.00143]	0.0737	
vanilla	seq2seq	1e+04	-0.0773	[-0.0999, -0.0547]	$< 10^{-4}$	*

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-3.226×10^{-4}	$[-6.50 \times 10^{-4}, 8.94 \times 10^{-5}]$	0.4495	-0.061	0.6573
adaptive	seq2seq	-0.01113	[-0.013, -0.00928]	$< 10^{-4}$	-0.573	0.0002
tropical	pointer	0.00174	$[-1.75 \times 10^{-4}, 0.00407]$	0.5713	0.145	0.2737
tropical	seq2seq	-0.05829	[-0.0658, -0.05]	$< 10^{-4}$	-0.528	0.0002
vanilla	pointer	2.845×10^{-4}	$[8.33 \times 10^{-5}, 5.04 \times 10^{-4}]$	0.3019	0.0971	0.4767
vanilla	seq2seq	-0.00877	[-0.0107, -0.00697]	$< 10^{-4}$	-0.356	0.0070

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.3023	[0.213, 0.392]	0.0014	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.307	[0.26, 0.354]	0.0007	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.3072	[0.256, 0.358]	0.0007	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.3124	[0.266, 0.359]	0.0007	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.33	[0.272, 0.388]	0.0008	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.3134	[0.277, 0.35]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.3272	[0.275, 0.379]	0.0007	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.3069	[0.258, 0.356]	0.0007	*

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.5112	[0.455, 0.567]	0.0002	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.5046	[0.4, 0.609]	0.0009	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.5273	[0.434, 0.621]	0.0008	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.532	[0.419, 0.645]	0.0009	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.5178	[0.425, 0.61]	0.0008	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.5063	[0.374, 0.639]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.3772	[0.321, 0.433]	0.0007	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.151	[0.0652, 0.237]	0.0081	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0138	[0.00713, 0.0205]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01253	[0.0058, 0.0193]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0127	[0.00656, 0.0188]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01213	[0.00529, 0.019]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01253	[0.00654, 0.0185]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01237	[0.00511, 0.0196]	0.0349	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01163	[0.00174, 0.0215]	0.0620	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0075	[6.22×10^{-4}, 0.0144]	0.0620	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.04863	[0.0391, 0.0581]	0.0018	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.05727	[0.0425, 0.0721]	0.0047	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.05707	[0.0417, 0.0724]	0.0050	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0548	[0.042, 0.0676]	0.0034	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.04957	[0.0407, 0.0584]	0.0015	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.05147	[0.0449, 0.058]	0.0004	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.03673	[0.03, 0.0435]	0.0016	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0499	[0.0312, 0.0686]	0.0160	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1031	[-0.113, -0.0934]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1082	[-0.113, -0.103]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1087	[-0.115, -0.102]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1072	[-0.111, -0.103]	$< 10^{-4}$	*

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1078	[-0.115, -0.101]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.1085	[-0.115, -0.102]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.07137	[-0.084, -0.0587]	0.0018	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.03183	[-0.0472, -0.0164]	0.0278	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.2772	[-0.399, -0.155]	0.0259	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.2611	[-0.349, -0.174]	0.0115	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.2845	[-0.377, -0.192]	0.0115	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.2841	[-0.375, -0.193]	0.0115	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.2585	[-0.385, -0.132]	0.0278	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.2622	[-0.41, -0.114]	0.0319	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.09627	[-0.178, -0.0141]	0.0626	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1665	[0.0489, 0.284]	0.0513	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.06827	[-0.081, -0.0556]	0.0020	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0635	[-0.0843, -0.0427]	0.0115	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.06433	[-0.0825, -0.0462]	0.0083	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.06457	[-0.0834, -0.0458]	0.0088	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.07077	[-0.0863, -0.0553]	0.0036	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.06937	[-0.0883, -0.0505]	0.0079	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.04627	[-0.0678, -0.0247]	0.0278	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01057	[-0.00355, 0.0247]	0.1063	

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.7683	[0.731, 0.805]	0.0473	5	0
adaptive	pointer	0.01	0.781	[0.755, 0.807]	0.0353	5	0
adaptive	pointer	0.1	0.7802	[0.754, 0.806]	0.0362	5	0
adaptive	pointer	1	0.7861	[0.76, 0.814]	0.0364	5	0
adaptive	pointer	10	0.7834	[0.753, 0.815]	0.041	5	0
adaptive	pointer	100	0.7813	[0.751, 0.807]	0.0359	5	0
adaptive	pointer	1000	0.7891	[0.756, 0.817]	0.0394	5	0
adaptive	pointer	1e+04	0.8043	[0.765, 0.837]	0.0459	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.8587	[0.845, 0.866]	0.015	5	0
adaptive	seq2seq	0.01	0.8608	[0.852, 0.866]	0.0104	5	0
adaptive	seq2seq	0.1	0.8606	[0.851, 0.866]	0.0108	5	0
adaptive	seq2seq	1	0.8608	[0.851, 0.866]	0.0104	5	0
adaptive	seq2seq	10	0.8594	[0.847, 0.866]	0.0135	5	0
adaptive	seq2seq	100	0.8596	[0.848, 0.866]	0.0131	5	0
adaptive	seq2seq	1000	0.8646	[0.863, 0.866]	0.00201	5	0
adaptive	seq2seq	1e+04	0.8657	[0.865, 0.866]	0.00108	5	0
tropical	pointer	0	0.8058	[0.783, 0.829]	0.0293	5	0
tropical	pointer	0.01	0.8092	[0.791, 0.828]	0.0237	5	0
tropical	pointer	0.1	0.778	[0.756, 0.797]	0.0259	5	0
tropical	pointer	1	0.7955	[0.785, 0.807]	0.014	5	0
tropical	pointer	10	0.7875	[0.776, 0.797]	0.0132	5	0
tropical	pointer	100	0.8125	[0.796, 0.837]	0.0281	5	0
tropical	pointer	1000	0.785	[0.775, 0.794]	0.0126	5	0
tropical	pointer	1e+04	0.7925	[0.786, 0.799]	0.00889	5	0
tropical	seq2seq	0	0.8033	[0.78, 0.827]	0.03	5	0
tropical	seq2seq	0.01	0.8152	[0.798, 0.833]	0.0225	5	0
tropical	seq2seq	0.1	0.8071	[0.788, 0.825]	0.0233	5	0
tropical	seq2seq	1	0.8125	[0.783, 0.841]	0.0364	5	0
tropical	seq2seq	10	0.8199	[0.788, 0.844]	0.0378	5	0
tropical	seq2seq	100	0.7946	[0.774, 0.823]	0.0326	5	0
tropical	seq2seq	1000	0.7847	[0.763, 0.809]	0.0303	5	0
tropical	seq2seq	1e+04	0.7999	[0.767, 0.841]	0.0484	5	0
vanilla	pointer	0	0.7801	[0.724, 0.823]	0.0661	5	0
vanilla	pointer	0.01	0.761	[0.698, 0.824]	0.0841	5	0
vanilla	pointer	0.1	0.7618	[0.698, 0.825]	0.0847	5	0
vanilla	pointer	1	0.7612	[0.699, 0.822]	0.0836	5	0
vanilla	pointer	10	0.7637	[0.7, 0.826]	0.0841	5	0
vanilla	pointer	100	0.7657	[0.698, 0.826]	0.0826	5	0
vanilla	pointer	1000	0.7412	[0.672, 0.81]	0.0856	5	0
vanilla	pointer	1e+04	0.7083	[0.646, 0.774]	0.0829	5	0
vanilla	seq2seq	0	0.8542	[0.841, 0.865]	0.0157	5	0
vanilla	seq2seq	0.01	0.8572	[0.845, 0.865]	0.0143	5	0
vanilla	seq2seq	0.1	0.8569	[0.843, 0.865]	0.015	5	0
vanilla	seq2seq	1	0.8575	[0.844, 0.865]	0.0144	5	0
vanilla	seq2seq	10	0.857	[0.844, 0.865]	0.0144	5	0
vanilla	seq2seq	100	0.8586	[0.845, 0.866]	0.0151	5	0
vanilla	seq2seq	1000	0.8624	[0.857, 0.866]	0.0058	5	0
vanilla	seq2seq	1e+04	0.8667	[0.865, 0.869]	0.00245	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.01279	[-0.0569, 0.0825]	0.9950
adaptive	pointer	0.1	0.01197	[-0.0578, 0.0817]	0.9966
adaptive	pointer	1	0.01782	[-0.0519, 0.0875]	0.9691
adaptive	pointer	10	0.01514	[-0.0546, 0.0849]	0.9869
adaptive	pointer	100	0.01303	[-0.0567, 0.0828]	0.9944
adaptive	pointer	1000	0.02084	[-0.0489, 0.0906]	0.9344
adaptive	pointer	1e+04	0.03602	[-0.0337, 0.106]	0.5690
adaptive	seq2seq	0.01	0.002033	[-0.0166, 0.0207]	0.9998
adaptive	seq2seq	0.1	0.001867	[-0.0168, 0.0205]	0.9999

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	seq2seq	1	0.002033	[-0.0166, 0.0207]	0.9998
adaptive	seq2seq	10	6.730×10^{-4}	[-0.018, 0.0193]	1.0000
adaptive	seq2seq	100	8.625×10^{-4}	[-0.0178, 0.0195]	1.0000
adaptive	seq2seq	1000	0.005888	[-0.0128, 0.0246]	0.9169
adaptive	seq2seq	1e+04	0.006912	[-0.0118, 0.0256]	0.8418
tropical	pointer	0.01	0.003463	[-0.033, 0.0399]	0.9999
tropical	pointer	0.1	-0.02782	[-0.0643, 0.00867]	0.1956
tropical	pointer	1	-0.0103	[-0.0468, 0.0262]	0.9497
tropical	pointer	10	-0.01825	[-0.0547, 0.0182]	0.6007
tropical	pointer	100	0.006763	[-0.0297, 0.0432]	0.9947
tropical	pointer	1000	-0.02078	[-0.0573, 0.0157]	0.4691
tropical	pointer	1e+04	-0.01329	[-0.0498, 0.0232]	0.8513
tropical	seq2seq	0.01	0.01181	[-0.0469, 0.0705]	0.9914
tropical	seq2seq	0.1	0.00371	[-0.055, 0.0624]	1.0000
tropical	seq2seq	1	0.00915	[-0.0495, 0.0678]	0.9981
tropical	seq2seq	10	0.01654	[-0.0421, 0.0752]	0.9500
tropical	seq2seq	100	-0.008729	[-0.0674, 0.0499]	0.9986
tropical	seq2seq	1000	-0.0186	[-0.0773, 0.0401]	0.9149
tropical	seq2seq	1e+04	-0.003446	[-0.0621, 0.0552]	1.0000
vanilla	pointer	0.01	-0.0191	[-0.162, 0.124]	0.9993
vanilla	pointer	0.1	-0.01832	[-0.161, 0.125]	0.9995
vanilla	pointer	1	-0.01887	[-0.162, 0.124]	0.9993
vanilla	pointer	10	-0.0164	[-0.16, 0.127]	0.9997
vanilla	pointer	100	-0.01439	[-0.157, 0.129]	0.9999
vanilla	pointer	1000	-0.03887	[-0.182, 0.104]	0.9582
vanilla	pointer	1e+04	-0.07179	[-0.215, 0.0713]	0.5981
vanilla	seq2seq	0.01	0.00293	[-0.0198, 0.0257]	0.9994
vanilla	seq2seq	0.1	0.002629	[-0.0201, 0.0254]	0.9997
vanilla	seq2seq	1	0.003264	[-0.0195, 0.026]	0.9989
vanilla	seq2seq	10	0.002737	[-0.02, 0.0255]	0.9996
vanilla	seq2seq	100	0.004337	[-0.0184, 0.0271]	0.9937
vanilla	seq2seq	1000	0.008207	[-0.0146, 0.031]	0.8568
vanilla	seq2seq	1e+04	0.01251	[-0.0103, 0.0353]	0.5064

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.002951	$[8.05 \times 10^{-4}, 0.00537]$	0.3401	0.139	0.3073
adaptive	seq2seq	7.681×10^{-4}	$[4.12 \times 10^{-7}, 0.0023]$	0.3362	0.12	0.3833
tropical	pointer	-6.826×10^{-4}	$[-0.00225, 5.55 \times 10^{-4}]$	0.7097	-0.0438	0.7485
tropical	seq2seq	-0.003867	[-0.00978, 0.00199]	0.1696	-0.211	0.1082
vanilla	pointer	-0.006953	[-0.00873, -0.00517]	0.3025	-0.109	0.4299
vanilla	seq2seq	0.001463	$[1.49 \times 10^{-4}, 0.00332]$	0.1512	0.225	0.0892

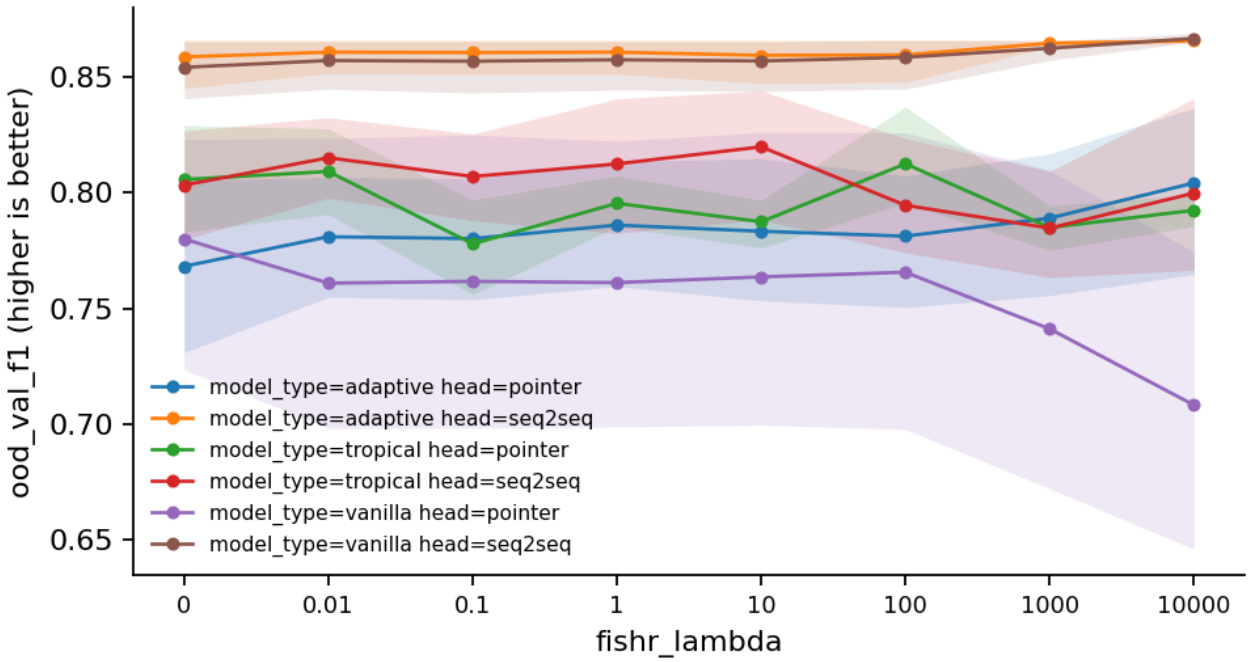


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02568	[-0.0432, 0.0945]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.04824	[-0.0515, 0.148]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01618	[-0.0991, 0.131]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.03425	[-0.0612, 0.13]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02384	[-0.084, 0.132]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04683	[-0.0436, 0.137]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.04377	[-0.0627, 0.15]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.08418	[-0.0129, 0.181]	0.6640
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.05089	[-0.092, -0.00972]	0.3177
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.042	[-0.0705, -0.0135]	0.1948
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.04981	[-0.0814, -0.0182]	0.1757
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.045	[-0.0924, 0.00237]	0.5772
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.03709	[-0.0927, 0.0185]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.06395	[-0.104, -0.0236]	0.1757
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0777	[-0.118, -0.0377]	0.0917
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.06684	[-0.127, -0.00622]	0.4137
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.01183	[-0.0989, 0.0752]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.02006	[-0.0887, 0.129]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.01846	[-0.0931, 0.13]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02486	[-0.0866, 0.136]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01971	[-0.0935, 0.133]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01559	[-0.0959, 0.127]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.04788	[-0.0838, 0.18]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.09597	[-0.0281, 0.22]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.004512	[-0.00746, 0.0165]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.003615	[-0.00262, 0.00985]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.00375	[-0.00279, 0.0103]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.003281	[-0.00255, 0.00911]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.002448	[-0.00229, 0.00718]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.001037	[-0.00152, 0.0036]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.002193	[-0.00283, 0.00721]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.001083	[-0.00539, 0.00323]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.09049	[-0.142, -0.0387]	0.1837
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.07973	[-0.122, -0.037]	0.1585
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.08038	[-0.125, -0.036]	0.1683
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.0747	[-0.119, -0.0301]	0.1991
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.07602	[-0.125, -0.0274]	0.2320
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.07832	[-0.125, -0.0318]	0.1991
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.07553	[-0.126, -0.0253]	0.2382
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.06138	[-0.119, -0.00354]	0.5458
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.002424	[-0.0359, 0.0407]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.005927	[-0.0318, 0.02]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.0291	[-0.0694, 0.0112]	0.8085
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01703	[-0.0608, 0.0267]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.03236	[-0.083, 0.0182]	0.9026
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01792	[0.00206, 0.0338]	0.5243
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	2.474×10^{-4}	[-0.0505, 0.051]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.007416	[-0.0676, 0.0527]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.07415	[-0.142, -0.00608]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.09617	[-0.188, -0.00449]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.09509	[-0.187, -0.00333]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.09628	[-0.186, -0.00618]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.09329	[-0.185, -0.00174]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.09287	[-0.18, -0.00552]	0.5458
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1212	[-0.222, -0.0206]	0.4598
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1584	[-0.262, -0.0548]	0.2382

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1009	[0.0546, 0.147]	0.0609	5	0
adaptive	pointer	0.01	0.1114	[0.0784, 0.146]	0.0441	5	0
adaptive	pointer	0.1	0.1102	[0.0774, 0.143]	0.0433	5	0
adaptive	pointer	1	0.1175	[0.0833, 0.154]	0.0456	5	0
adaptive	pointer	10	0.1134	[0.076, 0.157]	0.0538	5	0
adaptive	pointer	100	0.1091	[0.0761, 0.141]	0.041	5	0
adaptive	pointer	1000	0.1211	[0.084, 0.157]	0.0477	5	0
adaptive	pointer	1e+04	0.1447	[0.099, 0.189]	0.0601	5	0
adaptive	seq2seq	0	0.2177	[0.197, 0.23]	0.0233	5	0
adaptive	seq2seq	0.01	0.221	[0.207, 0.23]	0.0159	5	0
adaptive	seq2seq	0.1	0.2206	[0.206, 0.23]	0.017	5	0
adaptive	seq2seq	1	0.2211	[0.207, 0.23]	0.0159	5	0
adaptive	seq2seq	10	0.2185	[0.199, 0.23]	0.0216	5	0
adaptive	seq2seq	100	0.2191	[0.201, 0.23]	0.0204	5	0
adaptive	seq2seq	1000	0.2285	[0.226, 0.231]	0.00303	5	0
adaptive	seq2seq	1e+04	0.23	[0.226, 0.234]	0.0052	5	0
tropical	pointer	0	0.1401	[0.101, 0.18]	0.049	5	0
tropical	pointer	0.01	0.1498	[0.123, 0.18]	0.0355	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.09692	[0.0679, 0.119]	0.0343	5	0
tropical	pointer	1	0.1194	[0.0994, 0.139]	0.0256	5	0
tropical	pointer	10	0.1081	[0.0792, 0.127]	0.0322	5	0
tropical	pointer	100	0.1518	[0.125, 0.19]	0.0434	5	0
tropical	pointer	1000	0.1001	[0.0845, 0.116]	0.0214	5	0
tropical	pointer	1e+04	0.1206	[0.113, 0.129]	0.0106	5	0
tropical	seq2seq	0	0.1374	[0.104, 0.17]	0.0426	5	0
tropical	seq2seq	0.01	0.1554	[0.134, 0.18]	0.0298	5	0
tropical	seq2seq	0.1	0.1418	[0.113, 0.172]	0.0346	5	0
tropical	seq2seq	1	0.1482	[0.104, 0.192]	0.0573	5	0
tropical	seq2seq	10	0.1596	[0.11, 0.199]	0.0579	5	0
tropical	seq2seq	100	0.118	[0.089, 0.158]	0.0457	5	0
tropical	seq2seq	1000	0.1032	[0.0722, 0.136]	0.0421	5	0
tropical	seq2seq	1e+04	0.1323	[0.0697, 0.207]	0.0883	5	0
vanilla	pointer	0	0.1202	[0.0642, 0.167]	0.0658	5	0
vanilla	pointer	0.01	0.1159	[0.055, 0.177]	0.0798	5	0
vanilla	pointer	0.1	0.1171	[0.055, 0.179]	0.0807	5	0
vanilla	pointer	1	0.1167	[0.0573, 0.177]	0.0795	5	0
vanilla	pointer	10	0.118	[0.0553, 0.179]	0.0807	5	0
vanilla	pointer	100	0.1217	[0.0581, 0.184]	0.0822	5	0
vanilla	pointer	1000	0.09432	[0.0405, 0.155]	0.0701	5	0
vanilla	pointer	1e+04	0.06024	[0.0217, 0.107]	0.0534	5	0
vanilla	seq2seq	0	0.2095	[0.189, 0.228]	0.0248	5	0
vanilla	seq2seq	0.01	0.2138	[0.194, 0.227]	0.023	5	0
vanilla	seq2seq	0.1	0.213	[0.191, 0.227]	0.0244	5	0
vanilla	seq2seq	1	0.2147	[0.194, 0.228]	0.0233	5	0
vanilla	seq2seq	10	0.2134	[0.192, 0.228]	0.0234	5	0
vanilla	seq2seq	100	0.217	[0.195, 0.229]	0.0244	5	0
vanilla	seq2seq	1000	0.225	[0.22, 0.229]	0.00628	5	0
vanilla	seq2seq	1e+04	0.2335	[0.229, 0.24]	0.00746	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.01052	[-0.0769, 0.098]	0.9996
adaptive	pointer	0.1	0.00932	[-0.0781, 0.0968]	0.9998
adaptive	pointer	1	0.01662	[-0.0708, 0.104]	0.9938
adaptive	pointer	10	0.01248	[-0.075, 0.0999]	0.9989
adaptive	pointer	100	0.00826	[-0.0792, 0.0957]	0.9999
adaptive	pointer	1000	0.02024	[-0.0672, 0.108]	0.9815
adaptive	pointer	1e+04	0.04386	[-0.0436, 0.131]	0.5982
adaptive	seq2seq	0.01	0.00334	[-0.026, 0.0327]	0.9997
adaptive	seq2seq	0.1	0.00286	[-0.0265, 0.0322]	0.9999
adaptive	seq2seq	1	0.00338	[-0.0259, 0.0327]	0.9997
adaptive	seq2seq	10	7.600×10^{-4}	[-0.0286, 0.0301]	1.0000
adaptive	seq2seq	100	0.00142	[-0.0279, 0.0307]	1.0000
adaptive	seq2seq	1000	0.01078	[-0.0185, 0.0401]	0.8456
adaptive	seq2seq	1e+04	0.01234	[-0.017, 0.0417]	0.7541
tropical	pointer	0.01	0.00966	[-0.0489, 0.0682]	0.9973
tropical	pointer	0.1	-0.04318	[-0.102, 0.0153]	0.2217
tropical	pointer	1	-0.0207	[-0.0792, 0.0378]	0.8669
tropical	pointer	10	-0.03196	[-0.0905, 0.0266]	0.5125
tropical	pointer	100	0.0117	[-0.0468, 0.0702]	0.9917

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	-0.04002	[-0.0985, 0.0185]	0.2879
tropical	pointer	1e+04	-0.0195	[-0.078, 0.039]	0.8953
tropical	seq2seq	0.01	0.01806	[-0.0739, 0.11]	0.9926
tropical	seq2seq	0.1	0.00446	[-0.0875, 0.0965]	1.0000
tropical	seq2seq	1	0.01082	[-0.0812, 0.103]	0.9997
tropical	seq2seq	10	0.02224	[-0.0698, 0.114]	0.9767
tropical	seq2seq	100	-0.01936	[-0.111, 0.0726]	0.9890
tropical	seq2seq	1000	-0.03412	[-0.126, 0.0579]	0.8408
tropical	seq2seq	1e+04	-0.00502	[-0.097, 0.087]	1.0000
vanilla	pointer	0.01	-0.00432	[-0.135, 0.126]	1.0000
vanilla	pointer	0.1	-0.00304	[-0.133, 0.127]	1.0000
vanilla	pointer	1	-0.00346	[-0.134, 0.127]	1.0000
vanilla	pointer	10	-0.0022	[-0.133, 0.128]	1.0000
vanilla	pointer	100	0.00152	[-0.129, 0.132]	1.0000
vanilla	pointer	1000	-0.02586	[-0.156, 0.105]	0.9921
vanilla	pointer	1e+04	-0.05994	[-0.19, 0.0705]	0.6803
vanilla	seq2seq	0.01	0.00434	[-0.0323, 0.041]	0.9997
vanilla	seq2seq	0.1	0.00348	[-0.0332, 0.0401]	0.9999
vanilla	seq2seq	1	0.00516	[-0.0315, 0.0418]	0.9990
vanilla	seq2seq	10	0.0039	[-0.0327, 0.0405]	0.9998
vanilla	seq2seq	100	0.00746	[-0.0292, 0.0441]	0.9908
vanilla	seq2seq	1000	0.01548	[-0.0212, 0.0521]	0.7513
vanilla	seq2seq	1e+04	0.02398	[-0.0127, 0.0606]	0.3293

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.004054	$[9.45 \times 10^{-4}, 0.0071]$	0.2974	0.101	0.4543
adaptive	seq2seq	0.00146	$[7.14 \times 10^{-7}, 0.00437]$	0.2502	0.158	0.2452
tropical	pointer	-0.001741	$[-0.00433, 5.10 \times 10^{-4}]$	0.5614	-0.0876	0.5251
tropical	seq2seq	-0.006306	[-0.0167, 0.00478]	0.1592	-0.217	0.0980
vanilla	pointer	-0.007411	[-0.0103, -0.00361]	0.2278	-0.145	0.2857
vanilla	seq2seq	0.003044	$[5.10 \times 10^{-4}, 0.00627]$	0.0693	0.383	0.0032

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01992	[-0.0532, 0.0931]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0339	[-0.063, 0.131]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.02022	[-0.135, 0.095]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.00268	[-0.0942, 0.0995]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.00984	[-0.127, 0.107]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0301	[-0.0611, 0.121]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.00576	[-0.0796, 0.0911]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.06036	[0.00246, 0.118]	0.4881
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.07214	[-0.132, -0.0122]	0.3451
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.05842	[-0.0984, -0.0184]	0.2158
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.07116	[-0.122, -0.0201]	0.2338
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.06648	[-0.144, 0.0112]	0.6871
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0538	[-0.142, 0.0344]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.09896	[-0.157, -0.0406]	0.1390
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.1217	[-0.176, -0.0678]	0.0529
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.1011	[-0.213, 0.0108]	0.6609
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.0193	[-0.11, 0.071]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00446	[-0.112, 0.103]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00694	[-0.116, 0.102]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	7.800×10^{-4}	[-0.112, 0.114]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00462	[-0.123, 0.114]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.01256	[-0.124, 0.099]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0268	[-0.0953, 0.149]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0845	[-0.0144, 0.183]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.0082	[-0.0131, 0.0295]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.0072	[-0.00497, 0.0194]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.00758	[-0.00513, 0.0203]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.00642	[-0.0048, 0.0176]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.00506	[-0.00476, 0.0149]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00216	[-0.00295, 0.00727]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0035	[-0.00502, 0.012]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00344	[-0.0174, 0.0106]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1168	[-0.178, -0.056]	0.1190
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1096	[-0.161, -0.0586]	0.0888

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1104	[-0.161, -0.0594]	0.0888
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1036	[-0.158, -0.0489]	0.1190
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1051	[-0.167, -0.0433]	0.1557
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.11	[-0.162, -0.0576]	0.0903
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.1074	[-0.168, -0.0464]	0.1455
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.0853	[-0.163, -0.00803]	0.3510
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.00274	[-0.0659, 0.0713]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.00566	[-0.0528, 0.0414]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.0449	[-0.1, 0.0105]	0.6126
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.02878	[-0.104, 0.0469]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.05146	[-0.136, 0.0327]	0.9894
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.0338	[0.00928, 0.0583]	0.2614
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.00316	[-0.0727, 0.0664]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01174	[-0.116, 0.0923]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.08932	[-0.152, -0.0271]	0.2448
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.09798	[-0.18, -0.0162]	0.3510
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.09584	[-0.178, -0.0138]	0.3510
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.09794	[-0.178, -0.0184]	0.3492
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.09542	[-0.178, -0.0131]	0.3510
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.09526	[-0.176, -0.015]	0.3510
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1307	[-0.211, -0.0502]	0.1718
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1732	[-0.244, -0.103]	0.0582

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8397	[0.837, 0.842]	0.00351	5	0
adaptive	pointer	0.01	0.8392	[0.837, 0.842]	0.00374	5	0
adaptive	pointer	0.1	0.8391	[0.837, 0.842]	0.00301	5	0
adaptive	pointer	1	0.8389	[0.837, 0.841]	0.00299	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.8392	[0.837, 0.842]	0.00332	5	0
adaptive	pointer	100	0.8389	[0.837, 0.842]	0.00304	5	0
adaptive	pointer	1000	0.8396	[0.836, 0.843]	0.00459	5	0
adaptive	pointer	1e+04	0.8414	[0.837, 0.846]	0.00541	5	0
adaptive	seq2seq	0	0.86	[0.857, 0.862]	0.00332	5	0
adaptive	seq2seq	0.01	0.8602	[0.858, 0.862]	0.00304	5	0
adaptive	seq2seq	0.1	0.8603	[0.858, 0.863]	0.00308	5	0
adaptive	seq2seq	1	0.8603	[0.857, 0.863]	0.00322	5	0
adaptive	seq2seq	10	0.8597	[0.857, 0.862]	0.00288	5	0
adaptive	seq2seq	100	0.86	[0.857, 0.862]	0.00345	5	0
adaptive	seq2seq	1000	0.8611	[0.858, 0.863]	0.0035	5	0
adaptive	seq2seq	1e+04	0.8638	[0.861, 0.866]	0.00327	5	0
tropical	pointer	0	0.8531	[0.852, 0.854]	0.00168	5	0
tropical	pointer	0.01	0.8535	[0.851, 0.856]	0.00354	5	0
tropical	pointer	0.1	0.8528	[0.85, 0.856]	0.00376	5	0
tropical	pointer	1	0.8526	[0.85, 0.855]	0.00322	5	0
tropical	pointer	10	0.8524	[0.848, 0.855]	0.00488	5	0
tropical	pointer	100	0.8539	[0.851, 0.856]	0.00324	5	0
tropical	pointer	1000	0.8523	[0.849, 0.855]	0.00394	5	0
tropical	pointer	1e+04	0.8524	[0.849, 0.856]	0.00433	5	0
tropical	seq2seq	0	0.8567	[0.855, 0.858]	0.0017	5	0
tropical	seq2seq	0.01	0.8583	[0.857, 0.859]	0.00133	5	0
tropical	seq2seq	0.1	0.8583	[0.857, 0.86]	0.00209	5	0
tropical	seq2seq	1	0.8586	[0.858, 0.86]	0.00144	5	0
tropical	seq2seq	10	0.8581	[0.857, 0.86]	0.00159	5	0
tropical	seq2seq	100	0.8573	[0.856, 0.859]	0.00204	5	0
tropical	seq2seq	1000	0.8552	[0.853, 0.857]	0.00293	5	0
tropical	seq2seq	1e+04	0.8502	[0.846, 0.854]	0.00501	5	0
vanilla	pointer	0	0.8336	[0.83, 0.837]	0.00436	5	0
vanilla	pointer	0.01	0.8322	[0.829, 0.835]	0.0035	5	0
vanilla	pointer	0.1	0.8326	[0.83, 0.835]	0.0032	5	0
vanilla	pointer	1	0.8323	[0.83, 0.835]	0.00316	5	0
vanilla	pointer	10	0.8324	[0.83, 0.835]	0.00369	5	0
vanilla	pointer	100	0.832	[0.829, 0.835]	0.00386	5	0
vanilla	pointer	1000	0.8314	[0.829, 0.834]	0.00361	5	0
vanilla	pointer	1e+04	0.8263	[0.822, 0.83]	0.00523	5	0
vanilla	seq2seq	0	0.8594	[0.856, 0.863]	0.00425	5	0
vanilla	seq2seq	0.01	0.8599	[0.858, 0.862]	0.00314	5	0
vanilla	seq2seq	0.1	0.8599	[0.857, 0.862]	0.00318	5	0
vanilla	seq2seq	1	0.8598	[0.857, 0.863]	0.00365	5	0
vanilla	seq2seq	10	0.8599	[0.857, 0.863]	0.00339	5	0
vanilla	seq2seq	100	0.86	[0.858, 0.862]	0.00285	5	0
vanilla	seq2seq	1000	0.8608	[0.859, 0.863]	0.00265	5	0
vanilla	seq2seq	1e+04	0.8611	[0.859, 0.864]	0.00292	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-4.947×10^{-4}	[-0.00712, 0.00613]	1.0000
adaptive	pointer	0.1	-6.112×10^{-4}	[-0.00723, 0.00601]	0.9999
adaptive	pointer	1	-7.778×10^{-4}	[-0.0074, 0.00584]	0.9997
adaptive	pointer	10	-4.867×10^{-4}	[-0.00711, 0.00613]	1.0000
adaptive	pointer	100	-7.837×10^{-4}	[-0.00741, 0.00584]	0.9997

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	1000	-7.232×10^{-5}	[-0.00669, 0.00655]	1.0000	
adaptive	pointer	1e+04	0.001732	[-0.00489, 0.00835]	0.9652	
adaptive	seq2seq	0.01	1.970×10^{-4}	[-0.00544, 0.00583]	1.0000	
adaptive	seq2seq	0.1	3.185×10^{-4}	[-0.00532, 0.00595]	1.0000	
adaptive	seq2seq	1	3.422×10^{-4}	[-0.00529, 0.00598]	1.0000	
adaptive	seq2seq	10	-3.037×10^{-4}	[-0.00594, 0.00533]	1.0000	
adaptive	seq2seq	100	-3.890×10^{-5}	[-0.00567, 0.0056]	1.0000	
adaptive	seq2seq	1000	0.00113	[-0.00451, 0.00677]	0.9916	
adaptive	seq2seq	1e+04	0.003835	[-0.0018, 0.00947]	0.2924	
tropical	pointer	0.01	3.599×10^{-4}	[-0.00607, 0.00679]	1.0000	
tropical	pointer	0.1	-3.332×10^{-4}	[-0.00676, 0.0061]	1.0000	
tropical	pointer	1	-4.734×10^{-4}	[-0.0069, 0.00596]	1.0000	
tropical	pointer	10	-7.188×10^{-4}	[-0.00715, 0.00571]	0.9998	
tropical	pointer	100	7.601×10^{-4}	[-0.00567, 0.00719]	0.9997	
tropical	pointer	1000	-7.920×10^{-4}	[-0.00722, 0.00564]	0.9996	
tropical	pointer	1e+04	-6.456×10^{-4}	[-0.00708, 0.00578]	0.9999	
tropical	seq2seq	0.01	0.001534	[-0.0029, 0.00597]	0.8777	
tropical	seq2seq	0.1	0.001617	[-0.00282, 0.00605]	0.8506	
tropical	seq2seq	1	0.001847	[-0.00259, 0.00628]	0.7619	
tropical	seq2seq	10	0.001411	[-0.00302, 0.00584]	0.9135	
tropical	seq2seq	100	6.040×10^{-4}	[-0.00383, 0.00504]	0.9992	
tropical	seq2seq	1000	-0.001485	[-0.00592, 0.00295]	0.8930	
tropical	seq2seq	1e+04	-0.006489	[-0.0109, -0.00206]	0.0019	*
vanilla	pointer	0.01	-0.001337	[-0.00811, 0.00544]	0.9923	
vanilla	pointer	0.1	-0.001003	[-0.00778, 0.00577]	0.9986	
vanilla	pointer	1	-0.001278	[-0.00805, 0.0055]	0.9941	
vanilla	pointer	10	-0.001154	[-0.00793, 0.00562]	0.9968	
vanilla	pointer	100	-0.001562	[-0.00834, 0.00521]	0.9819	
vanilla	pointer	1000	-0.002217	[-0.00899, 0.00456]	0.9028	
vanilla	pointer	1e+04	-0.007333	[-0.0141, -5.58×10^{-4}]	0.0295	*
vanilla	seq2seq	0.01	5.279×10^{-4}	[-0.00522, 0.00627]	0.9999	
vanilla	seq2seq	0.1	5.147×10^{-4}	[-0.00523, 0.00626]	0.9999	
vanilla	seq2seq	1	4.149×10^{-4}	[-0.00533, 0.00616]	1.0000	
vanilla	seq2seq	10	5.644×10^{-4}	[-0.00518, 0.00631]	0.9999	
vanilla	seq2seq	100	5.986×10^{-4}	[-0.00515, 0.00634]	0.9999	
vanilla	seq2seq	1000	0.001431	[-0.00431, 0.00718]	0.9728	
vanilla	seq2seq	1e+04	0.001757	[-0.00399, 0.0075]	0.9274	

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.769×10^{-4}	$[-2.79 \times 10^{-5}, 4.91 \times 10^{-4}]$	0.3672	0.0971	0.4699
adaptive	seq2seq	4.342×10^{-4}	$[2.62 \times 10^{-5}, 7.88 \times 10^{-4}]$	0.1101	0.2	0.1420
tropical	pointer	-9.645×10^{-5}	$[-5.91 \times 10^{-4}, 3.84 \times 10^{-4}]$	0.7543	0.00952	0.9464
tropical	seq2seq	-0.001126	[-0.00156, -7.70×10^{-4}]	$< 10^{-4}$	-0.478	0.0004
vanilla	pointer	-7.393×10^{-4}	[-0.00111, -4.16×10^{-4}]	0.0288	-0.253	0.0564
vanilla	seq2seq	2.037×10^{-4}	$[-6.49 \times 10^{-5}, 4.64 \times 10^{-4}]$	0.4111	0.135	0.3069

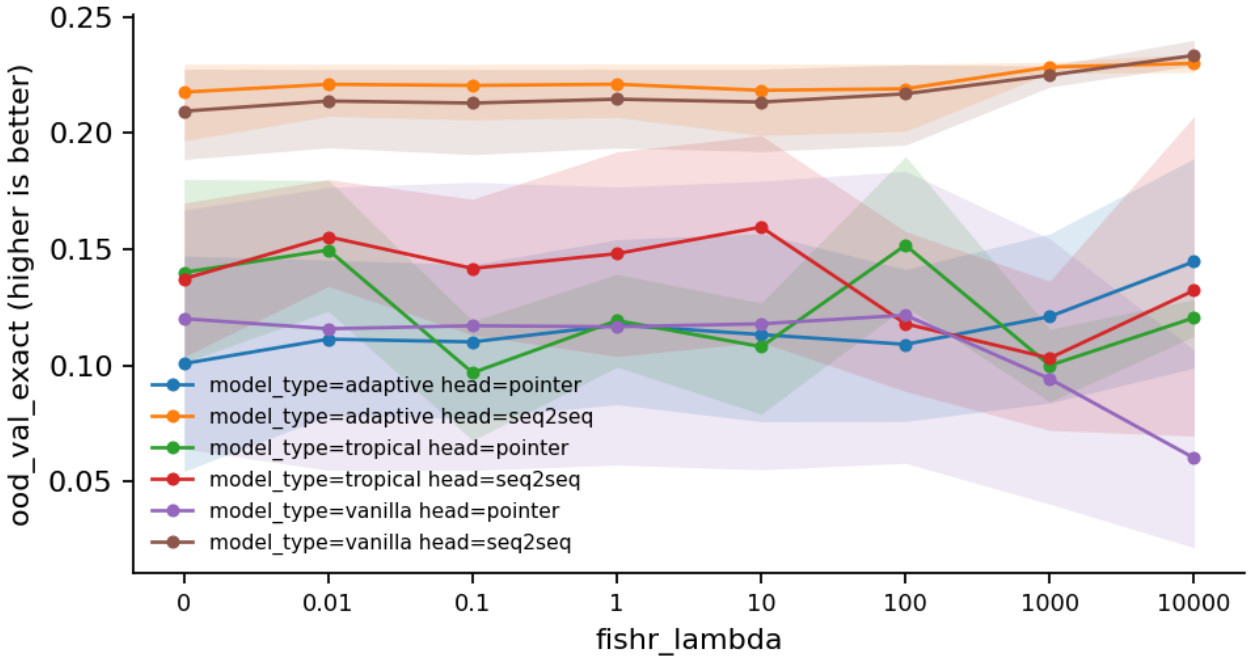


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

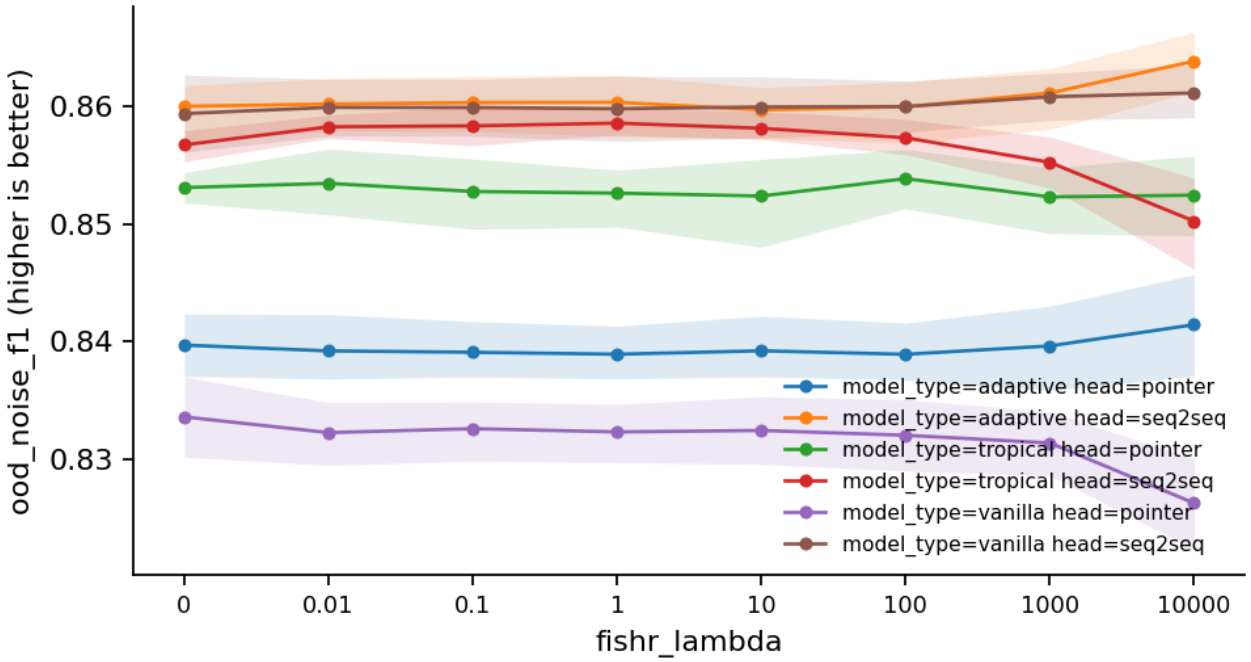


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01951	[0.0143, 0.0248]	0.0079	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.02121	[0.0136, 0.0288]	0.0194	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.02018	[0.0127, 0.0277]	0.0209	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.02031	[0.0134, 0.0272]	0.0183	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01994	[0.0116, 0.0283]	0.0272	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.02183	[0.0144, 0.0293]	0.0183	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.02093	[0.0124, 0.0295]	0.0272	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0262	[0.0146, 0.0378]	0.0301	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.002659	[-0.00778, 0.00246]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.001652	[-0.00575, 0.00245]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.001557	[-0.00619, 0.00308]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.001227	[-0.0058, 0.00335]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.001812	[-0.0054, 0.00178]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.002654	[-0.00666, 0.00135]	0.8395	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.005575	[-0.0105, -6.79×10^{-4}]	0.2389	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.0109	[-0.0181, -0.00368]	0.1105	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.006109	[-0.00339, 0.0156]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.006952	[-0.00142, 0.0153]	0.8836	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.006501	[-3.61×10^{-4}, 0.0134]	0.8463	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.006609	[-2.91×10^{-4}, 0.0135]	0.8463	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.006776	[-0.0013, 0.0149]	0.8836	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.006888	[-0.00105, 0.0148]	0.8836	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.008254	[-7.70×10^{-4}, 0.0173]	0.8463	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01517	[0.0034, 0.0269]	0.3710	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	6.208×10^{-4}	[-0.0037, 0.00495]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	2.899×10^{-4}	[-0.00247, 0.00305]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	4.246×10^{-4}	[-0.00261, 0.00346]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	5.482×10^{-4}	[-0.00284, 0.00394]	1.0000	

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-2.473×10^{-4}	[-0.00359, 0.0031]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-1.670×10^{-5}	[-0.00347, 0.00343]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	3.200×10^{-4}	[-0.00348, 0.00412]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0027	[-0.00273, 0.00812]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.02031	[-0.0253, -0.0153]	0.0076	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.021	[-0.0275, -0.0145]	0.0105	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.02124	[-0.0274, -0.0151]	0.0099	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.02143	[-0.0278, -0.015]	0.0105	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0205	[-0.0262, -0.0148]	0.0091	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02106	[-0.0274, -0.0147]	0.0105	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.02152	[-0.0295, -0.0136]	0.0154	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.02242	[-0.0305, -0.0143]	0.0154	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.003634	[-0.00678, -4.84×10^{-4}]	0.1987	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.004808	[-0.0105, 8.93×10^{-4}]	0.3170	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.005583	[-0.0102, -9.66×10^{-4}]	0.1987	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.005954	[-0.0104, -0.0015]	0.1654	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.005764	[-0.0111, -3.93×10^{-4}]	0.2037	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.003477	[-0.00789, 9.33×10^{-4}]	0.3170	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.002941	[-0.0098, 0.00392]	0.5992	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.00221	[-0.00822, 0.0126]	0.5992	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.0258	[-0.0343, -0.0173]	0.0121	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.02767	[-0.0348, -0.0205]	0.0086	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02732	[-0.0343, -0.0203]	0.0086	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.02749	[-0.0349, -0.0201]	0.0089	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02752	[-0.035, -0.0201]	0.0089	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.02796	[-0.0352, -0.0207]	0.0086	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.02945	[-0.0364, -0.0225]	0.0067	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.03489	[-0.0419, -0.0279]	0.0039	*

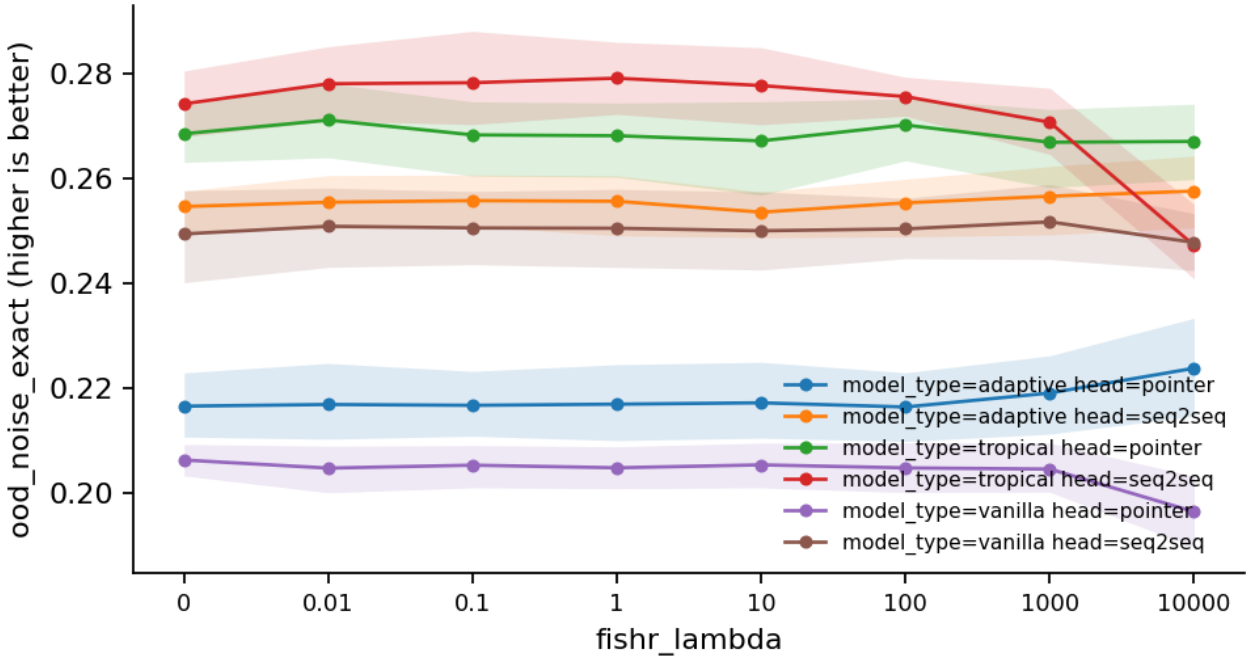


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.2166	[0.211, 0.223]	0.0077	5	0
adaptive	pointer	0.01	0.217	[0.21, 0.225]	0.0094	5	0
adaptive	pointer	0.1	0.2168	[0.211, 0.223]	0.00821	5	0
adaptive	pointer	1	0.217	[0.21, 0.225]	0.00914	5	0
adaptive	pointer	10	0.2173	[0.211, 0.225]	0.00949	5	0
adaptive	pointer	100	0.2165	[0.21, 0.223]	0.00827	5	0
adaptive	pointer	1000	0.2191	[0.211, 0.226]	0.00947	5	0
adaptive	pointer	1e+04	0.2239	[0.215, 0.233]	0.0127	5	0
adaptive	seq2seq	0	0.2547	[0.25, 0.258]	0.00511	5	0
adaptive	seq2seq	0.01	0.2555	[0.25, 0.261]	0.00695	5	0
adaptive	seq2seq	0.1	0.2558	[0.251, 0.261]	0.00656	5	0
adaptive	seq2seq	1	0.2557	[0.249, 0.261]	0.00685	5	0
adaptive	seq2seq	10	0.2536	[0.249, 0.257]	0.00583	5	0
adaptive	seq2seq	100	0.2554	[0.249, 0.26]	0.00722	5	0
adaptive	seq2seq	1000	0.2566	[0.249, 0.262]	0.00874	5	0
adaptive	seq2seq	1e+04	0.2576	[0.251, 0.264]	0.00896	5	0
tropical	pointer	0	0.2685	[0.263, 0.274]	0.007	5	0
tropical	pointer	0.01	0.2711	[0.264, 0.278]	0.00916	5	0
tropical	pointer	0.1	0.2683	[0.261, 0.275]	0.00941	5	0
tropical	pointer	1	0.2681	[0.26, 0.274]	0.00942	5	0
tropical	pointer	10	0.2672	[0.257, 0.275]	0.0116	5	0
tropical	pointer	100	0.2702	[0.263, 0.275]	0.00783	5	0
tropical	pointer	1000	0.2669	[0.258, 0.273]	0.0104	5	0
tropical	pointer	1e+04	0.2671	[0.26, 0.274]	0.00943	5	0
tropical	seq2seq	0	0.2742	[0.268, 0.28]	0.00836	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.278	[0.271, 0.285]	0.00923	5	0
tropical	seq2seq	0.1	0.2782	[0.27, 0.288]	0.0113	5	0
tropical	seq2seq	1	0.2791	[0.272, 0.286]	0.00887	5	0
tropical	seq2seq	10	0.2777	[0.27, 0.285]	0.00973	5	0
tropical	seq2seq	100	0.2756	[0.272, 0.279]	0.00473	5	0
tropical	seq2seq	1000	0.2707	[0.265, 0.277]	0.0081	5	0
tropical	seq2seq	1e+04	0.2473	[0.241, 0.255]	0.00922	5	0
vanilla	pointer	0	0.2064	[0.203, 0.209]	0.00383	5	0
vanilla	pointer	0.01	0.2048	[0.2, 0.209]	0.00567	5	0
vanilla	pointer	0.1	0.2054	[0.201, 0.209]	0.0054	5	0
vanilla	pointer	1	0.2049	[0.201, 0.209]	0.00509	5	0
vanilla	pointer	10	0.2055	[0.201, 0.21]	0.0054	5	0
vanilla	pointer	100	0.2049	[0.2, 0.21]	0.00628	5	0
vanilla	pointer	1000	0.2047	[0.2, 0.21]	0.00623	5	0
vanilla	pointer	1e+04	0.1965	[0.19, 0.203]	0.00862	5	0
vanilla	seq2seq	0	0.2494	[0.24, 0.258]	0.0111	5	0
vanilla	seq2seq	0.01	0.2509	[0.243, 0.258]	0.00967	5	0
vanilla	seq2seq	0.1	0.2506	[0.244, 0.258]	0.00937	5	0
vanilla	seq2seq	1	0.2505	[0.243, 0.258]	0.0101	5	0
vanilla	seq2seq	10	0.25	[0.243, 0.257]	0.00986	5	0
vanilla	seq2seq	100	0.2504	[0.245, 0.256]	0.00747	5	0
vanilla	seq2seq	1000	0.2517	[0.245, 0.259]	0.00947	5	0
vanilla	seq2seq	1e+04	0.2478	[0.243, 0.253]	0.00683	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	3.200×10^{-4}	[-0.0161, 0.0167]	1.0000
adaptive	pointer	0.1	1.600×10^{-4}	[-0.0163, 0.0166]	1.0000
adaptive	pointer	1	3.800×10^{-4}	[-0.016, 0.0168]	1.0000
adaptive	pointer	10	6.400×10^{-4}	[-0.0158, 0.0171]	1.0000
adaptive	pointer	100	-1.600×10^{-4}	[-0.0166, 0.0163]	1.0000
adaptive	pointer	1000	0.0025	[-0.0139, 0.0189]	0.9984
adaptive	pointer	1e+04	0.00722	[-0.0092, 0.0236]	0.7190
adaptive	seq2seq	0.01	8.200×10^{-4}	[-0.0116, 0.0133]	1.0000
adaptive	seq2seq	0.1	0.00112	[-0.0113, 0.0136]	0.9999
adaptive	seq2seq	1	0.001	[-0.0115, 0.0135]	1.0000
adaptive	seq2seq	10	-0.00108	[-0.0135, 0.0114]	1.0000
adaptive	seq2seq	100	7.000×10^{-4}	[-0.0118, 0.0132]	1.0000
adaptive	seq2seq	1000	0.00196	[-0.0105, 0.0144]	0.9980
adaptive	seq2seq	1e+04	0.00294	[-0.00952, 0.0154]	0.9795
tropical	pointer	0.01	0.0026	[-0.0138, 0.019]	0.9979
tropical	pointer	0.1	-2.200×10^{-4}	[-0.0166, 0.0162]	1.0000
tropical	pointer	1	-3.800×10^{-4}	[-0.0168, 0.016]	1.0000
tropical	pointer	10	-0.00136	[-0.0177, 0.015]	1.0000
tropical	pointer	100	0.00166	[-0.0147, 0.018]	0.9999
tropical	pointer	1000	-0.00162	[-0.018, 0.0148]	0.9999
tropical	pointer	1e+04	-0.00146	[-0.0178, 0.0149]	0.9999
tropical	seq2seq	0.01	0.00382	[-0.0117, 0.0193]	0.9742
tropical	seq2seq	0.1	0.00402	[-0.0115, 0.0195]	0.9666
tropical	seq2seq	1	0.00488	[-0.0106, 0.0204]	0.9173
tropical	seq2seq	10	0.00348	[-0.012, 0.019]	0.9842
tropical	seq2seq	100	0.00138	[-0.0141, 0.0169]	0.9999

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	seq2seq	1000	-0.0035	[-0.019, 0.012]	0.9837	
tropical	seq2seq	1e+04	-0.02696	[-0.0424, -0.0115]	0.0002	*
vanilla	pointer	0.01	-0.00154	[-0.0119, 0.00886]	0.9986	
vanilla	pointer	0.1	-9.800×10^{-4}	[-0.0114, 0.00942]	0.9999	
vanilla	pointer	1	-0.00148	[-0.0119, 0.00892]	0.9989	
vanilla	pointer	10	-9.200×10^{-4}	[-0.0113, 0.00948]	1.0000	
vanilla	pointer	100	-0.0015	[-0.0119, 0.0089]	0.9988	
vanilla	pointer	1000	-0.00172	[-0.0121, 0.00868]	0.9973	
vanilla	pointer	1e+04	-0.00984	[-0.0202, 5.60×10^{-4}]	0.0698	
vanilla	seq2seq	0.01	0.00146	[-0.0148, 0.0177]	0.9999	
vanilla	seq2seq	0.1	0.00114	[-0.0151, 0.0174]	1.0000	
vanilla	seq2seq	1	0.00108	[-0.0152, 0.0174]	1.0000	
vanilla	seq2seq	10	6.000×10^{-4}	[-0.0157, 0.0169]	1.0000	
vanilla	seq2seq	100	9.600×10^{-4}	[-0.0153, 0.0172]	1.0000	
vanilla	seq2seq	1000	0.0023	[-0.014, 0.0186]	0.9990	
vanilla	seq2seq	1e+04	-0.00162	[-0.0179, 0.0147]	0.9999	

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	8.871×10^{-4}	[4.91×10^{-4}, 0.00128]	0.2532	0.137	0.3001
adaptive	seq2seq	2.764×10^{-4}	[-5.94×10^{-4}, 0.00111]	0.6373	0.0781	0.5645
tropical	pointer	-4.621×10^{-4}	[-0.00152, 6.53×10^{-4}]	0.5474	-0.0667	0.6251
tropical	seq2seq	-0.00396	[-0.00426, -0.00367]	0.0002	-0.404	0.0024
vanilla	pointer	-9.429×10^{-4}	[-0.00145, -4.46×10^{-4}]	0.0804	-0.179	0.1814
vanilla	seq2seq	-2.514×10^{-4}	[-8.14×10^{-4}, 3.11×10^{-4}]	0.7251	-0.0514	0.6983

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.06214	[0.0518, 0.0725]	0.0012	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06628	[0.0514, 0.0812]	0.0032	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.0629	[0.0466, 0.0792]	0.0052	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.06324	[0.0495, 0.077]	0.0031	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.0617	[0.045, 0.0784]	0.0056	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0653	[0.0515, 0.0791]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.06224	[0.0445, 0.08]	0.0062	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.07052	[0.0484, 0.0927]	0.0081	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02478	[0.00521, 0.0444]	0.0550	

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02714	[0.0111, 0.0432]	0.0470	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02766	[0.00935, 0.046]	0.0550	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.02858	[0.0128, 0.0444]	0.0442	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02766	[0.0142, 0.0411]	0.0329	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.0252	[0.0145, 0.0359]	0.0228	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.01898	[0.00596, 0.032]	0.0550	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-5.600×10^{-4}	[-0.0116, 0.0105]	0.8946	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01026	[-0.00367, 0.0242]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01212	[-0.0038, 0.028]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0114	[-0.00132, 0.0241]	0.9471	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01212	[-0.00252, 0.0268]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01182	[-0.00425, 0.0279]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0116	[-0.00331, 0.0265]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01448	$[-1.94 \times 10^{-4}$, 0.0292]	0.7788	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02732	[0.00563, 0.049]	0.3996	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00522	[-0.00679, 0.0172]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00458	[-0.00261, 0.0118]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0052	[-0.00319, 0.0136]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.00514	[-0.00325, 0.0135]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.00354	[-0.00553, 0.0126]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00496	[-0.00261, 0.0125]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00488	[-0.00244, 0.0122]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00978	[-0.00283, 0.0224]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.03802	[-0.05, -0.026]	0.0219	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.03852	[-0.0556, -0.0215]	0.0397	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.03898	[-0.0552, -0.0228]	0.0393	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.03864	[-0.0557, -0.0216]	0.0397	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0363	[-0.052, -0.0206]	0.0397	*

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.03888	[-0.0553, -0.0225]	0.0393	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.03748	[-0.0567, -0.0183]	0.0563	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.03374	[-0.0546, -0.0129]	0.0983	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.0057	[-0.0145, 0.00309]	0.7306	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.00692	[-0.0244, 0.0106]	0.7306	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.00994	[-0.0263, 0.00643]	0.7306	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01096	[-0.0207, -0.00123]	0.2469	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01054	[-0.0187, -0.00235]	0.1868	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.00542	[-0.0151, 0.00421]	0.7306	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.00382	[-0.0161, 0.00843]	0.7306	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0198	[0.0015, 0.0381]	0.2469	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.04306	[-0.0595, -0.0266]	0.0304	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.04606	[-0.0628, -0.0293]	0.0301	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.04518	[-0.0619, -0.0285]	0.0301	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.04562	[-0.0619, -0.0293]	0.0298	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.04458	[-0.0601, -0.0291]	0.0282	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.04552	[-0.0599, -0.0312]	0.0219	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.04708	[-0.0643, -0.0298]	0.0301	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.05128	[-0.0676, -0.035]	0.0219	*

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9677	[0.967, 0.968]	8.79×10^{-4}	5	0
adaptive	pointer	0.01	0.9676	[0.967, 0.969]	0.00126	5	0
adaptive	pointer	0.1	0.9677	[0.967, 0.969]	0.00124	5	0
adaptive	pointer	1	0.9676	[0.967, 0.969]	0.00126	5	0
adaptive	pointer	10	0.9676	[0.967, 0.969]	0.00131	5	0
adaptive	pointer	100	0.9678	[0.967, 0.969]	0.00121	5	0
adaptive	pointer	1000	0.9679	[0.967, 0.969]	0.00118	5	0
adaptive	pointer	1e+04	0.9675	[0.966, 0.969]	0.00138	5	0
adaptive	seq2seq	0	0.9686	[0.968, 0.969]	4.46×10^{-4}	5	0
adaptive	seq2seq	0.01	0.969	[0.969, 0.969]	5.22×10^{-4}	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.9691	[0.969, 0.97]	6.92×10^{-4}	5	0
adaptive	seq2seq	1	0.969	[0.969, 0.969]	5.93×10^{-4}	5	0
adaptive	seq2seq	10	0.9691	[0.969, 0.97]	6.42×10^{-4}	5	0
adaptive	seq2seq	100	0.969	[0.968, 0.969]	6.36×10^{-4}	5	0
adaptive	seq2seq	1000	0.9653	[0.964, 0.966]	0.00108	5	0
adaptive	seq2seq	1e+04	0.9555	[0.955, 0.956]	6.37×10^{-4}	5	0
tropical	pointer	0	0.9839	[0.982, 0.985]	0.0019	5	0
tropical	pointer	0.01	0.9845	[0.984, 0.985]	0.00107	5	0
tropical	pointer	0.1	0.9845	[0.983, 0.986]	0.00141	5	0
tropical	pointer	1	0.9845	[0.984, 0.986]	0.00131	5	0
tropical	pointer	10	0.9847	[0.984, 0.986]	0.00133	5	0
tropical	pointer	100	0.9844	[0.983, 0.985]	0.00132	5	0
tropical	pointer	1000	0.9847	[0.983, 0.986]	0.00181	5	0
tropical	pointer	1e+04	0.9844	[0.984, 0.985]	0.00101	5	0
tropical	seq2seq	0	0.9851	[0.984, 0.986]	7.99×10^{-4}	5	0
tropical	seq2seq	0.01	0.9861	[0.985, 0.987]	0.00166	5	0
tropical	seq2seq	0.1	0.9864	[0.985, 0.987]	0.00193	5	0
tropical	seq2seq	1	0.9861	[0.984, 0.987]	0.00218	5	0
tropical	seq2seq	10	0.9863	[0.985, 0.988]	0.00195	5	0
tropical	seq2seq	100	0.9861	[0.983, 0.989]	0.00346	5	0
tropical	seq2seq	1000	0.9828	[0.98, 0.986]	0.00367	5	0
tropical	seq2seq	1e+04	0.9612	[0.954, 0.969]	0.0105	5	0
vanilla	pointer	0	0.9668	[0.966, 0.967]	9.30×10^{-4}	5	0
vanilla	pointer	0.01	0.9668	[0.967, 0.967]	3.28×10^{-4}	5	0
vanilla	pointer	0.1	0.9668	[0.967, 0.967]	3.22×10^{-4}	5	0
vanilla	pointer	1	0.9669	[0.967, 0.967]	2.74×10^{-4}	5	0
vanilla	pointer	10	0.9667	[0.966, 0.967]	4.80×10^{-4}	5	0
vanilla	pointer	100	0.9667	[0.967, 0.967]	2.41×10^{-4}	5	0
vanilla	pointer	1000	0.9668	[0.966, 0.967]	3.78×10^{-4}	5	0
vanilla	pointer	1e+04	0.9666	[0.966, 0.967]	4.26×10^{-4}	5	0
vanilla	seq2seq	0	0.9662	[0.965, 0.967]	0.00103	5	0
vanilla	seq2seq	0.01	0.9661	[0.965, 0.967]	8.43×10^{-4}	5	0
vanilla	seq2seq	0.1	0.9661	[0.965, 0.967]	8.03×10^{-4}	5	0
vanilla	seq2seq	1	0.9661	[0.966, 0.967]	7.14×10^{-4}	5	0
vanilla	seq2seq	10	0.9663	[0.966, 0.967]	8.70×10^{-4}	5	0
vanilla	seq2seq	100	0.966	[0.965, 0.967]	8.59×10^{-4}	5	0
vanilla	seq2seq	1000	0.9621	[0.961, 0.963]	8.63×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.9496	[0.948, 0.951]	0.0022	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-1.067×10^{-4}	[-0.00224, 0.00203]	1.0000
adaptive	pointer	0.1	-3.920×10^{-5}	[-0.00217, 0.00209]	1.0000
adaptive	pointer	1	-1.175×10^{-4}	[-0.00225, 0.00202]	1.0000
adaptive	pointer	10	-1.784×10^{-4}	[-0.00231, 0.00196]	1.0000
adaptive	pointer	100	1.124×10^{-5}	[-0.00212, 0.00215]	1.0000
adaptive	pointer	1000	1.420×10^{-4}	[-0.00199, 0.00228]	1.0000
adaptive	pointer	1e+04	-2.482×10^{-4}	[-0.00238, 0.00189]	0.9997
adaptive	seq2seq	0.01	4.753×10^{-4}	$[-7.11 \times 10^{-4}, 0.00166]$	0.7905
adaptive	seq2seq	0.1	5.553×10^{-4}	$[-6.31 \times 10^{-4}, 0.00174]$	0.6636
adaptive	seq2seq	1	4.621×10^{-4}	$[-7.24 \times 10^{-4}, 0.00165]$	0.8099
adaptive	seq2seq	10	5.189×10^{-4}	$[-6.67 \times 10^{-4}, 0.0017]$	0.7229

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	4.044×10^{-4}	$[-7.82 \times 10^{-4}, 0.00159]$	0.8848	
adaptive	seq2seq	1000	-0.003278	[-0.00446, -0.00209]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.01308	[-0.0143, -0.0119]	$< 10^{-4}$	*
tropical	pointer	0.01	5.964×10^{-4}	[-0.00189, 0.00309]	0.9778	
tropical	pointer	0.1	5.844×10^{-4}	[-0.00191, 0.00307]	0.9801	
tropical	pointer	1	6.078×10^{-4}	[-0.00188, 0.0031]	0.9755	
tropical	pointer	10	7.952×10^{-4}	[-0.00169, 0.00328]	0.9122	
tropical	pointer	100	4.887×10^{-4}	[-0.002, 0.00298]	0.9926	
tropical	pointer	1000	7.538×10^{-4}	[-0.00174, 0.00324]	0.9305	
tropical	pointer	1e+04	5.231×10^{-4}	[-0.00197, 0.00301]	0.9891	
tropical	seq2seq	0.01	0.001024	[-0.0066, 0.00865]	0.9993	
tropical	seq2seq	0.1	0.001302	[-0.00632, 0.00892]	0.9967	
tropical	seq2seq	1	0.001064	[-0.00656, 0.00869]	0.9991	
tropical	seq2seq	10	0.001258	[-0.00636, 0.00888]	0.9973	
tropical	seq2seq	100	0.001077	[-0.00655, 0.0087]	0.9990	
tropical	seq2seq	1000	-0.002307	[-0.00993, 0.00532]	0.9307	
tropical	seq2seq	1e+04	-0.02381	[-0.0314, -0.0162]	$< 10^{-4}$	*
vanilla	pointer	0.01	3.799×10^{-5}	$[-7.82 \times 10^{-4}, 8.58 \times 10^{-4}]$	1.0000	
vanilla	pointer	0.1	5.710×10^{-5}	$[-7.63 \times 10^{-4}, 8.77 \times 10^{-4}]$	1.0000	
vanilla	pointer	1	9.707×10^{-5}	$[-7.23 \times 10^{-4}, 9.17 \times 10^{-4}]$	0.9997	
vanilla	pointer	10	-3.534×10^{-5}	$[-8.56 \times 10^{-4}, 7.85 \times 10^{-4}]$	1.0000	
vanilla	pointer	100	-8.239×10^{-5}	$[-9.03 \times 10^{-4}, 7.38 \times 10^{-4}]$	0.9999	
vanilla	pointer	1000	-5.456×10^{-6}	$[-8.26 \times 10^{-4}, 8.15 \times 10^{-4}]$	1.0000	
vanilla	pointer	1e+04	-1.265×10^{-4}	$[-9.47 \times 10^{-4}, 6.94 \times 10^{-4}]$	0.9982	
vanilla	seq2seq	0.01	-9.163×10^{-5}	[-0.00204, 0.00186]	1.0000	
vanilla	seq2seq	0.1	-9.607×10^{-5}	[-0.00205, 0.00186]	1.0000	
vanilla	seq2seq	1	-9.605×10^{-5}	[-0.00205, 0.00186]	1.0000	
vanilla	seq2seq	10	9.630×10^{-5}	[-0.00186, 0.00205]	1.0000	
vanilla	seq2seq	100	-1.943×10^{-4}	[-0.00215, 0.00176]	0.9999	
vanilla	seq2seq	1000	-0.004044	[-0.006, -0.00209]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	-0.01656	[-0.0185, -0.0146]	$< 10^{-4}$	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.384×10^{-6}	$[-3.71 \times 10^{-5}, 4.19 \times 10^{-5}]$	0.9809	-0.0019	1.0000
adaptive	seq2seq	-0.001728	[-0.00184, -0.00163]	$< 10^{-4}$	-0.547	0.0002
tropical	pointer	-8.180×10^{-9}	$[-2.43 \times 10^{-5}, 1.99 \times 10^{-5}]$	0.9999	0.00952	0.9492
tropical	seq2seq	-0.002919	[-0.00391, -0.00187]	0.0001	-0.436	0.0006
vanilla	pointer	-2.850×10^{-5}	$[-6.86 \times 10^{-5}, 1.54 \times 10^{-5}]$	0.3194	-0.202	0.1328
vanilla	seq2seq	-0.00205	[-0.00233, -0.00182]	$< 10^{-4}$	-0.535	0.0002

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01716	[0.0141, 0.0202]	0.0004	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01772	[0.0163, 0.0192]	$< 10^{-4}$	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01769	[0.0157, 0.0197]	0.0002	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01767	[0.016, 0.0193]	0.0001	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01799	[0.016, 0.02]	0.0002	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01773	[0.016, 0.0194]	0.0001	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.01792	[0.0153, 0.0205]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01781	[0.0163, 0.0193]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.01887	[0.0172, 0.0206]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01999	[0.0172, 0.0227]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02027	[0.0172, 0.0234]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.02003	[0.0168, 0.0233]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02004	[0.017, 0.023]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.02015	[0.0155, 0.0248]	0.0005	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.02061	[0.0167, 0.0245]	0.0004	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01162	$[-3.70 \times 10^{-4}, 0.0236]$	0.0546	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	9.876×10^{-4}	$[-5.59 \times 10^{-4}, 0.00253]$	0.9051	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	8.428×10^{-4}	$[-7.15 \times 10^{-4}, 0.0024]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	8.913×10^{-4}	$[-7.30 \times 10^{-4}, 0.00251]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	7.730×10^{-4}	$[-8.96 \times 10^{-4}, 0.00244]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	8.445×10^{-4}	$[-8.63 \times 10^{-4}, 0.00255]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.001081	$[-3.84 \times 10^{-4}, 0.00255]$	0.7692	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.001135	$[-2.78 \times 10^{-4}, 0.00255]$	0.7173	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	8.658×10^{-4}	[-0.00112, 0.00285]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.002363	[0.0014, 0.00332]	0.0240	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00293	[0.00225, 0.00361]	0.0034	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.003015	[0.00247, 0.00356]	0.0016	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.002922	[0.0023, 0.00354]	0.0028	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.002786	[0.00247, 0.0031]	0.0003	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.002962	[0.0022, 0.00372]	0.0046	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.003129	[0.00242, 0.00384]	0.0033	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.005842	[0.00298, 0.00871]	0.0431	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-8.032×10^{-4}	[-0.00169, 8.46×10^{-5}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.001385	[-0.00309, 3.17×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.001398	[-0.00329, 4.99×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.001383	[-0.00322, 4.58×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0015	[-0.00307, 6.89×10^{-5}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.001196	[-0.00298, 5.91×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.002617	[4.59×10^{-4}, 0.00477]	0.5627	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01203	[0.00959, 0.0145]	0.0036	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.00114	[-0.00292, 6.41×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.001567	[-0.00344, 3.05×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.001858	[-0.00434, 6.28×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.001597	[-0.00429, 0.00109]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.001603	[-0.00412, 9.12×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.001729	[-0.00574, 0.00228]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.00192	[-0.0031, 0.00694]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0232	[0.00963, 0.0368]	0.1890	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	5.727×10^{-4}	[-0.00165, 0.0028]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	7.023×10^{-4}	[-5.89×10^{-4}, 0.00199]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	7.259×10^{-4}	[-5.50×10^{-4}, 0.002]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	7.658×10^{-4}	[-2.18×10^{-4}, 0.00175]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	4.411×10^{-4}	[-8.32×10^{-4}, 0.00171]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	6.846×10^{-4}	[-4.34×10^{-4}, 0.0018]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.004611	[0.00372, 0.0055]	0.0031	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01701	[0.0145, 0.0195]	0.0011	*

A.7 `id_exact`

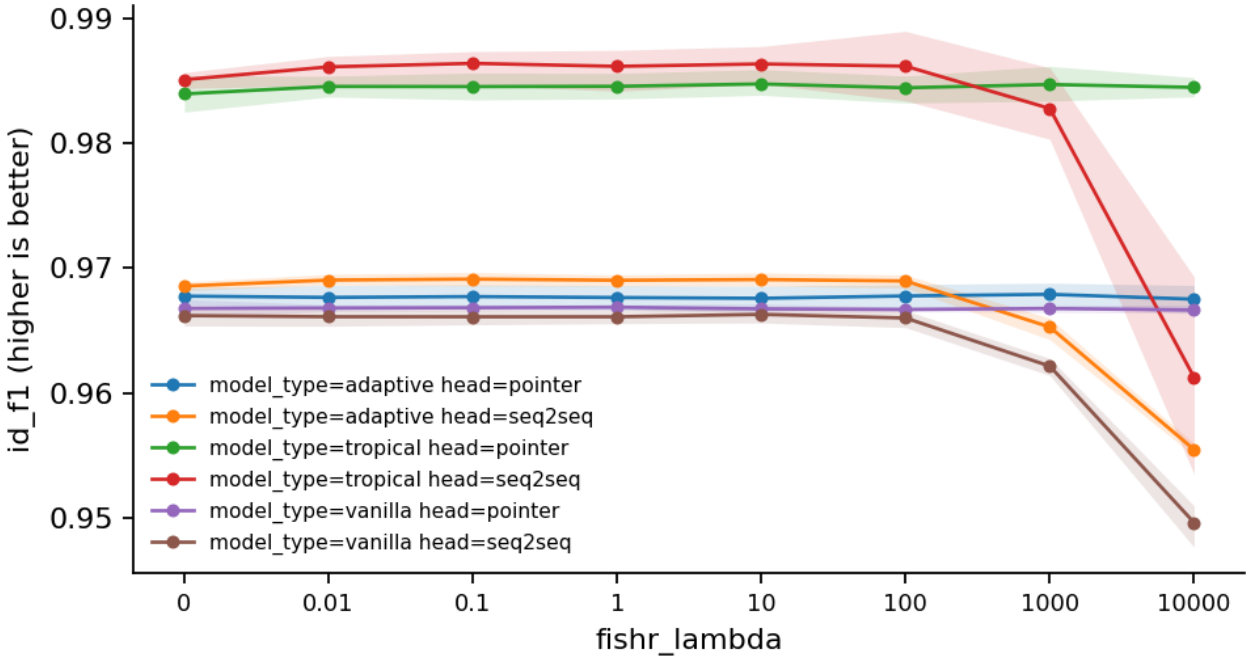


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

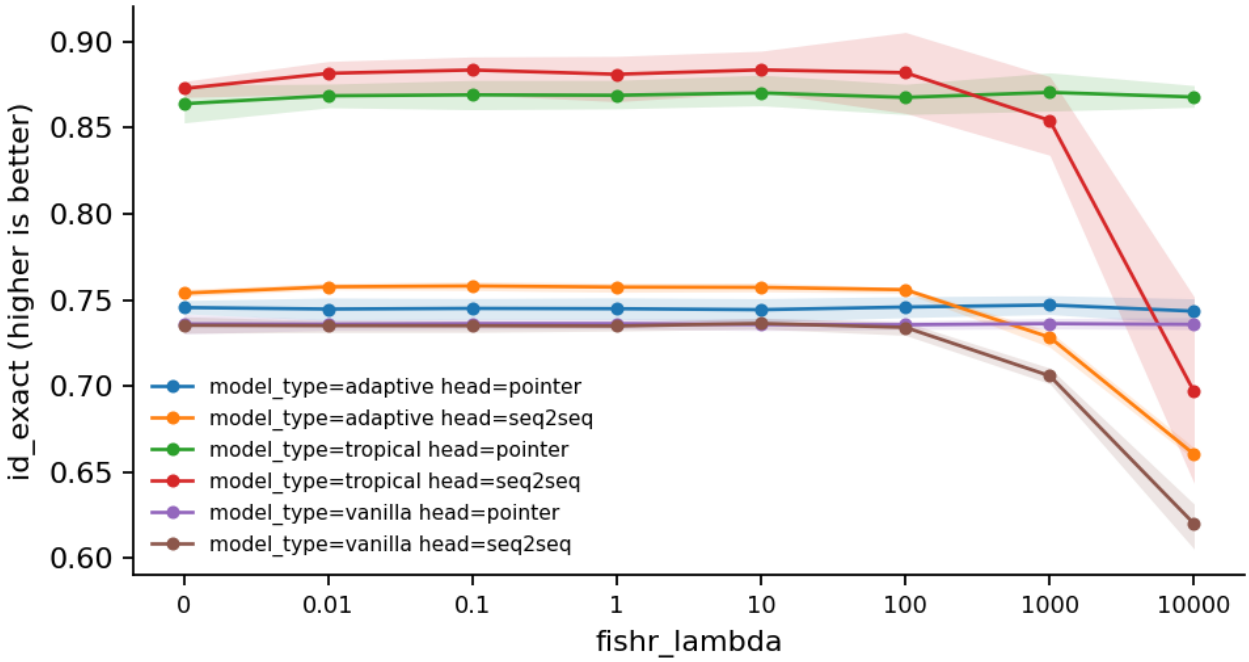


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.7455	[0.742, 0.75]	0.00499	5	0
adaptive	pointer	0.01	0.7447	[0.738, 0.751]	0.00832	5	0
adaptive	pointer	0.1	0.745	[0.739, 0.751]	0.00778	5	0
adaptive	pointer	1	0.7449	[0.739, 0.751]	0.00817	5	0
adaptive	pointer	10	0.7443	[0.737, 0.751]	0.00855	5	0
adaptive	pointer	100	0.7458	[0.74, 0.752]	0.00822	5	0
adaptive	pointer	1000	0.747	[0.742, 0.753]	0.00719	5	0
adaptive	pointer	1e+04	0.7434	[0.736, 0.751]	0.00936	5	0
adaptive	seq2seq	0	0.7539	[0.752, 0.756]	0.00256	5	0
adaptive	seq2seq	0.01	0.7576	[0.756, 0.76]	0.00235	5	0
adaptive	seq2seq	0.1	0.7581	[0.756, 0.761]	0.00329	5	0
adaptive	seq2seq	1	0.7574	[0.755, 0.76]	0.00334	5	0
adaptive	seq2seq	10	0.7573	[0.755, 0.76]	0.00326	5	0
adaptive	seq2seq	100	0.7559	[0.754, 0.757]	0.00212	5	0
adaptive	seq2seq	1000	0.7282	[0.723, 0.732]	0.00612	5	0
adaptive	seq2seq	1e+04	0.6602	[0.657, 0.664]	0.00442	5	0
tropical	pointer	0	0.8641	[0.853, 0.875]	0.0145	5	0
tropical	pointer	0.01	0.8687	[0.862, 0.876]	0.00872	5	0
tropical	pointer	0.1	0.8692	[0.861, 0.878]	0.0112	5	0
tropical	pointer	1	0.869	[0.861, 0.878]	0.0108	5	0
tropical	pointer	10	0.8704	[0.863, 0.881]	0.0118	5	0
tropical	pointer	100	0.8677	[0.858, 0.875]	0.011	5	0
tropical	pointer	1000	0.8707	[0.86, 0.882]	0.0147	5	0
tropical	pointer	1e+04	0.868	[0.862, 0.875]	0.00788	5	0
tropical	seq2seq	0	0.8729	[0.868, 0.877]	0.00606	5	0
tropical	seq2seq	0.01	0.8817	[0.87, 0.889]	0.0137	5	0
tropical	seq2seq	0.1	0.8837	[0.869, 0.891]	0.0158	5	0
tropical	seq2seq	1	0.8812	[0.865, 0.892]	0.0172	5	0
tropical	seq2seq	10	0.8838	[0.871, 0.895]	0.0154	5	0
tropical	seq2seq	100	0.8821	[0.859, 0.906]	0.0287	5	0
tropical	seq2seq	1000	0.8543	[0.834, 0.88]	0.0295	5	0
tropical	seq2seq	1e+04	0.6967	[0.643, 0.752]	0.0723	5	0
vanilla	pointer	0	0.736	[0.73, 0.742]	0.00749	5	0
vanilla	pointer	0.01	0.7359	[0.732, 0.739]	0.00398	5	0
vanilla	pointer	0.1	0.7362	[0.733, 0.739]	0.00384	5	0
vanilla	pointer	1	0.7362	[0.733, 0.739]	0.00376	5	0
vanilla	pointer	10	0.7356	[0.732, 0.739]	0.00463	5	0
vanilla	pointer	100	0.7355	[0.733, 0.738]	0.00324	5	0
vanilla	pointer	1000	0.7362	[0.733, 0.739]	0.00396	5	0
vanilla	pointer	1e+04	0.7357	[0.733, 0.74]	0.00461	5	0
vanilla	seq2seq	0	0.7352	[0.731, 0.74]	0.00612	5	0
vanilla	seq2seq	0.01	0.735	[0.731, 0.738]	0.00458	5	0
vanilla	seq2seq	0.1	0.735	[0.731, 0.738]	0.0044	5	0
vanilla	seq2seq	1	0.7348	[0.732, 0.737]	0.00338	5	0
vanilla	seq2seq	10	0.7363	[0.733, 0.739]	0.0042	5	0
vanilla	seq2seq	100	0.734	[0.729, 0.737]	0.005	5	0
vanilla	seq2seq	1000	0.7058	[0.701, 0.71]	0.00616	5	0
vanilla	seq2seq	1e+04	0.6198	[0.605, 0.631]	0.0176	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-8.800×10^{-4}	[-0.0147, 0.0129]	1.0000	
adaptive	pointer	0.1	-5.000×10^{-4}	[-0.0143, 0.0133]	1.0000	
adaptive	pointer	1	-6.800×10^{-4}	[-0.0145, 0.0131]	1.0000	
adaptive	pointer	10	-0.00126	[-0.0151, 0.0126]	0.9999	
adaptive	pointer	100	3.000×10^{-4}	[-0.0135, 0.0141]	1.0000	
adaptive	pointer	1000	0.0015	[-0.0123, 0.0153]	0.9998	
adaptive	pointer	1e+04	-0.00212	[-0.0159, 0.0117]	0.9983	
adaptive	seq2seq	0.01	0.00366	[-0.0027, 0.01]	0.4589	
adaptive	seq2seq	0.1	0.00414	[-0.00222, 0.0105]	0.3347	
adaptive	seq2seq	1	0.0035	[-0.00286, 0.00986]	0.5049	
adaptive	seq2seq	10	0.00334	[-0.00302, 0.0097]	0.5525	
adaptive	seq2seq	100	0.00198	[-0.00438, 0.00834]	0.9215	
adaptive	seq2seq	1000	-0.02572	[-0.0321, -0.0194]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.0937	[-0.1, -0.0873]	$< 10^{-4}$	*
tropical	pointer	0.01	0.0046	[-0.0156, 0.0248]	0.9830	
tropical	pointer	0.1	0.00514	[-0.0151, 0.0253]	0.9697	
tropical	pointer	1	0.0049	[-0.0153, 0.0251]	0.9763	
tropical	pointer	10	0.00628	[-0.0139, 0.0265]	0.9219	
tropical	pointer	100	0.00366	[-0.0165, 0.0239]	0.9953	
tropical	pointer	1000	0.00658	[-0.0136, 0.0268]	0.9046	
tropical	pointer	1e+04	0.00388	[-0.0163, 0.0241]	0.9934	
tropical	seq2seq	0.01	0.00884	[-0.0462, 0.0638]	0.9977	
tropical	seq2seq	0.1	0.01078	[-0.0442, 0.0658]	0.9926	
tropical	seq2seq	1	0.0083	[-0.0467, 0.0633]	0.9985	
tropical	seq2seq	10	0.01086	[-0.0441, 0.0659]	0.9923	
tropical	seq2seq	100	0.0092	[-0.0458, 0.0642]	0.9971	
tropical	seq2seq	1000	-0.01856	[-0.0736, 0.0364]	0.8896	
tropical	seq2seq	1e+04	-0.1762	[-0.231, -0.121]	$< 10^{-4}$	*
vanilla	pointer	0.01	-1.000×10^{-4}	[-0.00814, 0.00794]	1.0000	
vanilla	pointer	0.1	2.000×10^{-4}	[-0.00784, 0.00824]	1.0000	
vanilla	pointer	1	2.400×10^{-4}	[-0.0078, 0.00828]	1.0000	
vanilla	pointer	10	-3.400×10^{-4}	[-0.00838, 0.0077]	1.0000	
vanilla	pointer	100	-4.600×10^{-4}	[-0.0085, 0.00758]	1.0000	
vanilla	pointer	1000	2.200×10^{-4}	[-0.00782, 0.00826]	1.0000	
vanilla	pointer	1e+04	-2.600×10^{-4}	[-0.0083, 0.00778]	1.0000	
vanilla	seq2seq	0.01	-2.000×10^{-4}	[-0.0137, 0.0133]	1.0000	
vanilla	seq2seq	0.1	-2.600×10^{-4}	[-0.0138, 0.0133]	1.0000	
vanilla	seq2seq	1	-4.200×10^{-4}	[-0.014, 0.0131]	1.0000	
vanilla	seq2seq	10	0.00108	[-0.0125, 0.0146]	1.0000	
vanilla	seq2seq	100	-0.00124	[-0.0148, 0.0123]	0.9999	
vanilla	seq2seq	1000	-0.0294	[-0.0429, -0.0159]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	-0.1154	[-0.129, -0.102]	$< 10^{-4}$	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.500×10^{-5}	$[-2.57 \times 10^{-4}, 3.19 \times 10^{-4}]$	0.9451	0.0533	0.7009
adaptive	seq2seq	-0.01262	[-0.0135, -0.0118]	$< 10^{-4}$	-0.608	0.0002
tropical	pointer	-1.857×10^{-5}	$[-2.36 \times 10^{-4}, 1.28 \times 10^{-4}]$	0.9830	0.0381	0.7780
tropical	seq2seq	-0.02189	[-0.0288, -0.0144]	$< 10^{-4}$	-0.451	0.0004
vanilla	pointer	-4.071×10^{-5}	$[-4.24 \times 10^{-4}, 3.46 \times 10^{-4}]$	0.8977	-0.0286	0.8376

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.01446	[-0.0164, -0.0126]	$< 10^{-4}$	-0.543	0.0002

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1281	[0.105, 0.151]	0.0004	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1328	[0.121, 0.145]	$< 10^{-4}$	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.133	[0.118, 0.148]	0.0002	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1328	[0.119, 0.147]	0.0001	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1347	[0.117, 0.152]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1322	[0.119, 0.145]	0.0001	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1345	[0.115, 0.154]	0.0004	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1322	[0.121, 0.144]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1377	[0.128, 0.148]	$< 10^{-4}$	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1467	[0.127, 0.167]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1487	[0.126, 0.171]	0.0004	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1464	[0.123, 0.169]	0.0004	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1475	[0.126, 0.169]	0.0004	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1481	[0.111, 0.185]	0.0007	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1485	[0.118, 0.179]	0.0005	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.07686	[-0.00626, 0.16]	0.0622	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00956	[5.64×10^{-4}, 0.0186]	0.3356	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00878	[-0.00352, 0.0211]	0.4752	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00886	[-0.00244, 0.0202]	0.4752	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00864	[-0.00363, 0.0209]	0.4752	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00864	[-0.0042, 0.0215]	0.4752	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01032	[-0.00154, 0.0222]	0.4383	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01084	[1.54×10^{-4}, 0.0215]	0.3360	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0077	[-0.0061, 0.0215]	0.4752	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01872	[0.00993, 0.0275]	0.0410	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.02258	[0.018, 0.0272]	0.0022	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.02312	[0.0198, 0.0264]	0.0006	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.02264	[0.0184, 0.0269]	0.0017	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.02098	[0.0187, 0.0233]	0.0002	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02194	[0.0156, 0.0283]	0.0074	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0224	[0.0172, 0.0276]	0.0034	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.04044	[0.021, 0.0599]	0.0410	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.0084	[-0.0119, -0.00495]	0.0526	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.01294	[-0.0239, -0.00202]	0.5130	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01304	[-0.0247, -0.00136]	0.5802	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01258	[-0.0253, 1.67×10^{-4}]	0.7786	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.013	[-0.0231, -0.00287]	0.4228	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.01008	[-0.0215, 0.00137]	0.9933	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01882	[0.0047, 0.0329]	0.3955	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.08318	[0.0668, 0.0996]	0.0033	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.00882	[-0.0229, 0.0053]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.01306	[-0.0288, 0.00266]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01446	[-0.0334, 0.0045]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01222	[-0.0331, 0.00861]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.0134	[-0.0345, 0.00773]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.01436	[-0.0473, 0.0186]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01632	[-0.0238, 0.0564]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1713	[0.0775, 0.265]	0.1428	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	7.600×10^{-4}	[-0.0148, 0.0163]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	8.600×10^{-4}	[-0.00886, 0.0106]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00122	[-0.00809, 0.0105]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00142	[-0.00566, 0.0085]	1.0000	

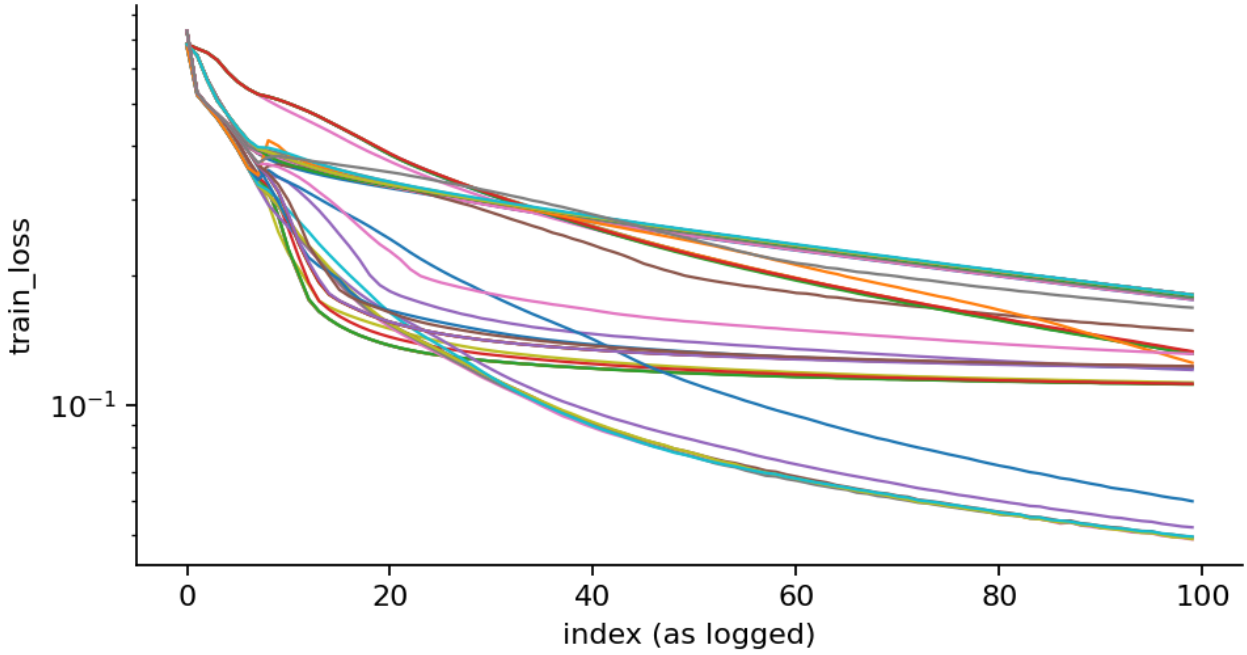


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-6.600×10^{-4}	[-0.00988, 0.00856]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.00154	[-0.00683, 0.00991]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.03038	[0.026, 0.0348]	0.0011	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1159	[0.0976, 0.134]	0.0014	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `aff15420a136`); factor levels are in Table 1.

```
task:
  name: ConvexHullDataset
  value_range:
  - 0
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
```

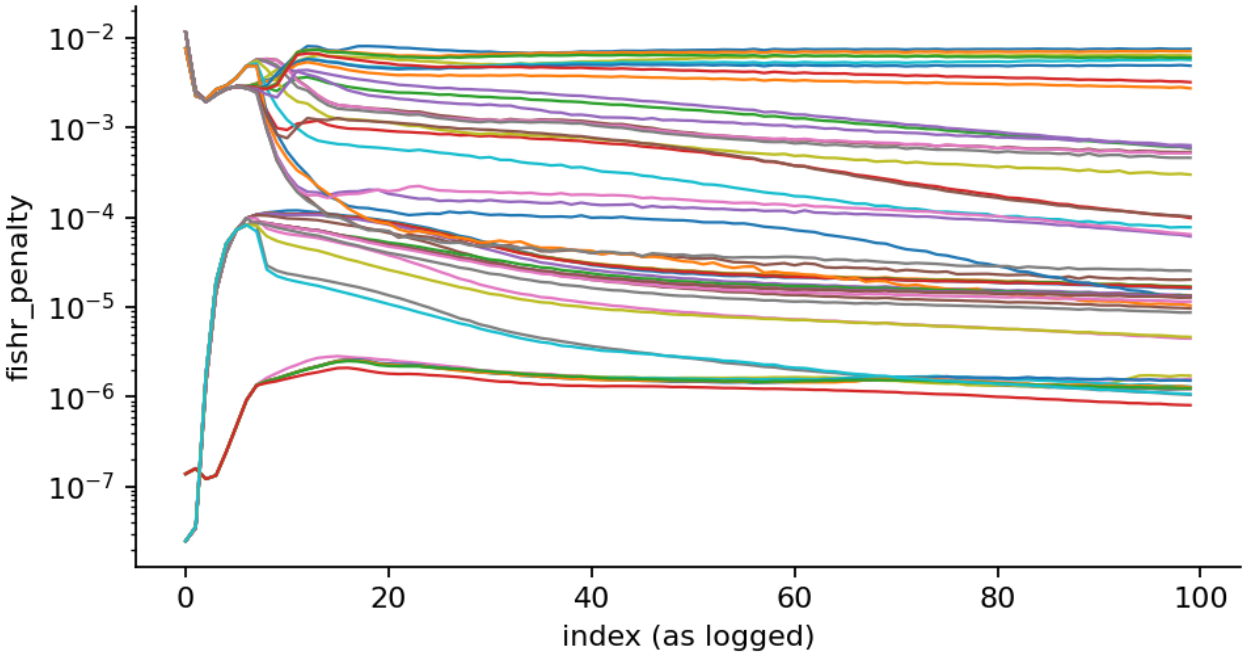


Figure 10: `fishr_penalty`, the mean over seeds per cell.

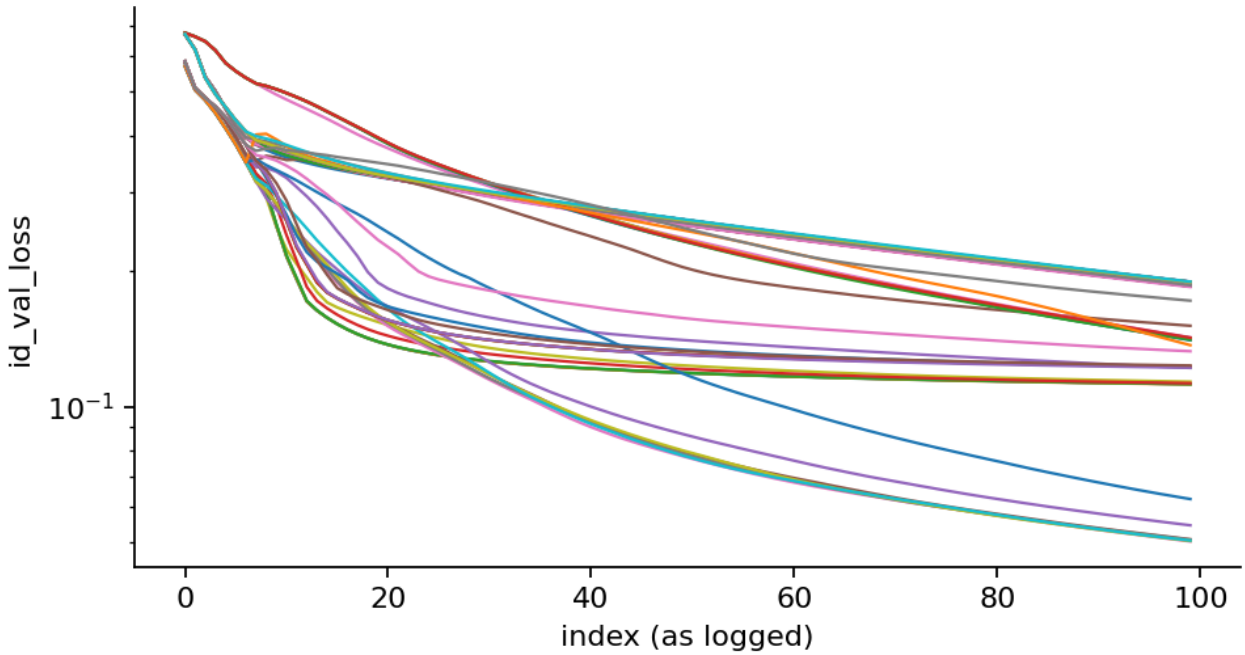


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/ConvexHull/results`.